

Full-length article

Sustainability reporting across religious-ethical traditions: A sociology of Worth perspective on UK and Italian water utilities

Gabriella D'Amore^{a,*}, Antonio D'Alessio^b, Alessandra Di Fraia^b, Fabio Serini^b,
Loris Landriani^b

^a Department of Law, University of Naples Parthenope, Italy

^b Department of Business and Economics, University of Naples Parthenope, Italy

ARTICLE INFO

Handling Editor: Janice A. Beecher

Keywords:

Water utilities

Sustainability reporting

Religion

Ethics

Sociology of worth (SOW)

Italy

UK

ABSTRACT

Purpose: This study investigates how religious-ethical traditions influence the justificatory frameworks employed in sustainability reporting. Focusing on water utilities in Italy and the United Kingdom, two contexts shaped respectively by Catholic and Protestant ethics, the research explores how different “worlds of value” are mobilised to construct legitimacy in sustainability narratives.

Design/methodology/approach: The study employs qualitative content analysis of 25 sustainability reports from Italian and UK water utilities. Using the Sociology of Worth as a theoretical lens, the analysis categorises disclosures into five normative “worlds of value” (Market, Industrial, Civic, Domestic, Green). The coding framework integrates sector-specific themes with religious-ethical orientations. Reports are evaluated based on frequency, descriptive depth, and strategic commitment of each value world.

Findings: The analysis reveals that UK companies prioritise Market, Industrial, and Green worlds, reflecting a Protestant ethic of individual accountability, performance, and rational stewardship. Italian companies, particularly publicly controlled utilities, emphasise Civic and Domestic values, aligning with Catholic principles of solidarity, subsidiarity, and relational legitimacy. While both contexts express environmental concern, the ethical anchoring of sustainability differs strategically and performance-based in the UK, symbolic and community-oriented in Italy.

Theoretical implications: The study contributes to the theorisation of sustainability reporting by integrating religious ethics into the Sociology of Worth. It demonstrates that sustainability disclosures are not culturally neutral, but embedded in normative infrastructures shaped by historical, institutional, and ethical contexts. The findings call for a rethinking of materiality as a culturally contingent construct and propose a more dialogical approach to standardisation in non-financial reporting.

Practical implications: For organisations, the study highlights the importance of culturally resonant ethical narratives in enhancing the legitimacy of sustainability communication. For regulators and standard-setters, it underscores the limits of one-size-fits-all approaches and the need for flexibility in disclosure frameworks. For stakeholders, the research offers tools to interpret the ethical assumptions underlying corporate sustainability claims.

Originality/value: This is the first study to systematically compare Catholic and Protestant ethical influences on the narrative construction of sustainability reporting using the Sociology of Worth. It expands the analytical scope of ethical and cultural perspectives in accounting research and offers a novel interpretive framework for evaluating the normative legitimacy of non-financial disclosure practices.

1. Introduction

In the context of an increasingly globalized and interconnected world, contemporary organisations are being urged to confront complex

and interrelated socio-environmental challenges such as climate change, poverty, inequality, and biodiversity loss (Bebbington and Larrinaga-González, 2008; O'Dwyer and Unerman, 2007). These challenges have redefined the boundaries of corporate responsibility, placing growing

* Corresponding author.

E-mail addresses: gabriella.damore@uniparthenope.it (G. D'Amore), loris.landriani@uniparthenope.it (L. Landriani).

<https://doi.org/10.1016/j.jup.2026.102211>

Received 19 September 2025; Received in revised form 9 April 2026; Accepted 13 April 2026

Available online 28 April 2026

0957-1787/© 2026 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

pressure on organisations to account not only to shareholders but to a broader range of stakeholders, including employees, consumers, local communities, and civil society (Freeman, 2023). Within this evolving landscape, sustainability reporting has emerged as a key mechanism for enhancing transparency, demonstrating legitimacy, and disclosing organisational commitments to sustainable development (Baldini et al., 2018; De Villiers and Marques, 2016).

Over the past decade, a growing body of literature has argued that sustainability reporting should not be understood merely as a technical or regulatory compliance tool, but as a cultural and ethical practice through which organisations construct meaning and articulate their value orientations (Hahn et al., 2010; Torelli, 2021). Reports are increasingly conceived as discursive artefacts, where narratives are strategically shaped to resonate with the moral expectations and cultural sensibilities of stakeholder audiences (O'Dochartaigh, 2019; Tregidga and Milne, 2006). This narrative turn has drawn attention to the symbolic and justificatory dimensions of sustainability disclosure, highlighting its role in shaping perceptions of legitimacy, identity, and purpose.

However, this narrative and symbolic potential also entails risks. In the absence of a robust ethical foundation, sustainability reporting may devolve into a performative or reputational exercise, masking unethical practices or fostering cognitive dissonance among stakeholders (Aji and Sutikno, 2015; Fassin et al., 2011). The instrumentalisation of ethics for strategic ends, commonly labelled as greenwashing or impression management, undermines the emancipatory potential of sustainability discourse, reducing accountability to a rhetorical device (Siano et al., 2017; Torelli et al., 2020).

To address these concerns, recent scholarship has called for a renewed focus on the ethical underpinnings of sustainability, positioning moral reasoning and normative values as critical foundations for credible and transformative reporting (Brunelli and Di Carlo, 2020; Torelli, 2021). Within this ethical reorientation, religiosity, conceived as a historically embedded system of beliefs, moral norms, and practices, has gained attention as a potentially influential cultural force shaping organisational behaviour and communication (Guiso et al., 2006; Liao et al., 2019). Religious traditions provide distinct ethical worldviews that inform how concepts such as justice, stewardship, responsibility, and transparency are interpreted and enacted (Brammer et al., 2007; Terzani et al., 2025; Terzani and Turzo, 2021). Nonetheless, despite the centrality of cultural and moral systems in institutional logics theory, the specific role of religious traditions has been systematically under-explored. Prior work has tended to treat religiosity as a generic contextual force rather than as a set of differentiated ethical logics, thereby leaving a significant conceptual gap in understanding how distinct religious worldviews shape organisational meaning-making processes.

A growing body of research indicates that religiosity influences corporate sustainability practices, primarily through its role as an informal institutional pressure shaping expectations of responsible behaviour. Prior studies show that firms located in more religious communities tend to exhibit higher levels of ESG disclosure (Terzani and Turzo, 2021), are more likely to adopt sustainability reporting assurance (Terzani and Turzo, 2021), and generally provide more extensive CSR reporting as a response to religiously grounded social norms (Chantziaras et al., 2020). The contributions demonstrate that religion functions as a contextual force capable of constraining managerial behaviour and promoting forms of accountability beyond formal regulatory requirements. However, this literature remains limited in several crucial ways. First, religiosity is almost invariably conceptualised as an aggregated contextual indicator – typically measured at country or county level – without examining the distinct ethical traditions associated with specific religions. As a result, the moral foundations that underpin different religious frameworks remain theoretically implicit and empirically unexamined. Second, existing studies predominantly focus on the extent or credibility-enhancing mechanisms of sustainability

reporting (such as ESG scores or assurance adoption), rather than on the content, narratives, and value justifications through which organisations construct meaning and legitimacy in their disclosures. Third, although some scholars distinguish between Catholic and Protestant contexts (e.g., Terzani and Turzo, 2021), these distinctions are used merely as proxies for divergent social control mechanisms and do not explore how ethical traditions themselves shape organisational sustainability narratives. This highlights the need for a more explicit theorisation of religious traditions as carriers of distinctive moral logics capable of structuring justificatory practices within sustainability reporting.

This study addresses this gap by examining how ethical-religious traditions may shape the value orientations conveyed in sustainability narratives within specific institutional contexts. It builds on the call by Farooq et al. (2019) to broaden the analysis of interreligious differences in non-financial disclosure across diverse institutional and cultural settings. Adopting a comparative perspective, the research focuses on water utilities in two contrasting contexts: Italy, whose institutional and societal fabric has been historically shaped by Roman Catholicism, and the United Kingdom, long influenced by Protestant ethics. These traditions articulate distinct views of responsibility, authority, and social order, offering a meaningful foundation for analysing the ways in which sustainability is interpreted, legitimised, and communicated. This comparative approach not only responds to theoretical gaps but also provides policymakers with insights into how ethical diversity influences reporting practices, thereby informing the development of culturally responsive regulatory frameworks.

The utility sector provides an especially relevant empirical context. As providers of essential public goods, these organisations fulfil a distinct institutional role, operating at the intersection of environmental stewardship, social equity and democratic accountability (D'Amore et al., 2024; Traxler and Greiling, 2019). Their activities are characterised by the coexistence of positive and negative externalities, creating public value while simultaneously generating potential environmental and social harms (Moore, 1997; Pizzi et al., 2020). This dual nature amplifies societal expectations regarding transparency and responsible conduct, making stakeholder acceptance a critical intangible asset (Garde Sánchez et al., 2017). In this context, sustainability reporting functions not merely as a communication device, but as a central lever of accountability through which utilities justify their actions, disclose complex impacts, and demonstrate alignment with their public service mandate (Imperiale et al., 2023). Nevertheless, utilities have often been criticised for disclosures that lack transparency, coherence or ethical authenticity, raising concerns about symbolic compliance, greenwashing and impression management (Aji and Sutikno, 2015; Delmas and Burbano, 2011; Manes-Rossi and Nicolo', 2022). Understanding the ethical foundations of these narratives, therefore, holds direct policy relevance, particularly for regulators tasked with improving the quality, comparability and cultural legitimacy of sustainability disclosures in essential services.

These tensions reinforce the need for analytical approaches capable of uncovering the moral and justificatory foundations that underpin sustainability narratives (Torelli, 2021). Such foundations are often rooted in broader cultural and ethical traditions that shape organisational understandings of responsibility and legitimacy. In this perspective, religious-ethical principles represent a meaningful source of these orienting values. To explore the role of these principles within reporting practices, the study draws on the Sociology of Worth (SOW) developed by Boltanski et al. (2006) and later extended by Boltanski et al. (2006). This framework conceptualises social justification through a plurality of normative “worlds of value”, each grounded in distinct moral logics and principles. However, while SOW provides a powerful lens for analysing how organisations construct legitimacy, it leaves the cultural and ethical origins of these justificatory logics largely implicit (Jagd, 2011). Integrating Catholic and Protestant traditions therefore offers a crucial theoretical enrichment: it makes explicit the moral foundations that

shape why certain worlds of value become salient in specific contexts, rather than treating value orientations as culturally neutral or universally applicable. This integration allows the analysis to reveal how deeply rooted ethical worldviews guide the selection, prioritisation, and framing of sustainability themes in different national settings. Sustainability disclosures are thus interpreted as moral narratives that mobilise different ethical grammars depending on their religious and institutional context.

Accordingly, the study addresses the following research questions:

RQ1: Which worlds of value prevail in the sustainability narratives of water utilities?

RQ2: How are these worlds influenced by the ethical-religious traditions and institutional configurations of Catholic- and Protestant-influenced contexts?

By concentrating on the value-based framing of sustainability themes, the research offers a culturally situated interpretation of reporting practices, advancing the understanding of how moral values shape non-financial disclosure. The findings not only contribute to theoretical debates on the ethics of sustainability reporting but also provide practical insights into the cultural legitimacy of corporate communication in the utility sector. In doing so, the study highlights the regulatory and policy relevance of recognising ethical and institutional diversity in sustainability disclosure. By showing that narratives reflect deeper moral traditions, the research underscores the limits of uniform reporting requirements and points to the need for public authorities, regulators, and standard setters to develop frameworks that balance comparability with cultural responsiveness. This perspective informs ongoing debates on how emerging sustainability standards can support meaningful accountability in essential public services while remaining sensitive to the value orientations that shape organisational communication.

The structure of the paper is as follows. Section 2 outlines the theoretical background, integrating literature on religiosity as a source of ethical orientation, the Sociology of Worth, and sector-specific reporting practices. Section 3 presents the research methodology, detailing the comparative design, coding framework, and content analysis of sustainability reports. Section 4 presents the empirical findings, highlighting patterns of justificatory logics in UK and Italian reports. Section 5 develops discussion, linking the observed differences to broader ethical and institutional traditions. Section 6 elaborates the theoretical and practical implications of the study, acknowledging the study's limitations and suggesting directions for future research.

2. Theoretical background

2.1. Religiosity as a cultural source of ethical orientation

If ethics provides the normative framework through which sustainability reporting is constructed and interpreted, religion constitutes one of its most enduring and influential sources. Religiosity, understood as the collective internalisation of moral norms and value systems, contributes to shaping what is considered right, appropriate, or necessary to communicate in corporate disclosures (Chantziaras et al., 2020; Koehrsen, 2015). Even within formally secular corporate environments, religious traditions often act as latent cultural codes that inform accountability practices and influence the style and substance of sustainability reporting (Torelli, 2021).

In Europe, Catholicism and Protestantism represent the two dominant religious traditions, each offering distinct ethical frameworks and mechanisms of moral regulation. Their respective influence extends beyond theological doctrine, permeating broader cultural patterns, institutional expectations, and organisational behaviours, particularly in the realm of social responsibility and communication.

Catholic ethics emphasises community, solidarity, and the common

good, privileging relationality and institutional mediation. These principles are rooted in Aristotelian-Thomistic thought and have been systematised through the Catholic Social Teaching (CST), which provides a normative basis for addressing social and economic issues. Foundational documents such as *Rerum Novarum* (Leo XIII) and *Mater et Magistra* (John XXIII) conceive the firm as a human community with duties towards workers and society. Later encyclicals, including *Centesimus Annus* (John Paul II) and *Caritas in Veritate* (Benedict XVI), explicitly address the moral responsibilities of businesses, balancing profit with social needs and transparency. Pope Francis's *Laudato Si'* further expands this perspective by framing environmental stewardship as an ethical obligation tied to intergenerational justice and systemic well-being (Rousseau, 2017). According to Bruni (2018), these values express a "spirit of the South", shaped by Renaissance Humanism and the Catholic Counter-Reformation, which places relationality and cooperation at the core of economic and social life, even at the cost of individual advantage.

Protestant ethics, by contrast, is grounded in individual responsibility, moral autonomy, and self-regulation, rejecting institutional intermediation between the individual and the divine (Arruñada, 2010). The Reformation diminished the authority of the Church, replacing it with peer-based forms of social control within communities, reinforcing behavioural discipline and personal accountability. This moral structure gave rise to what Weber (2003) famously described as the Protestant ethic: a value system centred on hard work, thrift, discipline, and a rational approach to economic life. Protestant traditions prioritise personal initiative, education, and financial independence, viewing economic success not as a goal in itself, but as a by-product of a morally ordered life (Sherkat and Ellison, 2007). In environmental matters, Protestant communities have often promoted responsible resource management and individual behavioural change as means of addressing ecological degradation. Bruni (2018) identifies these values with the "spirit of the North", characterised by discipline, economic rationality, and the construction of a capitalist order based on individual performance and systemic efficiency.

These divergent ethical traditions shape not only private moral reasoning but also collective attitudes toward accountability and transparency. Catholicism tends to foster a narrative style grounded in solidarity, care for the vulnerable, and systemic balance, elements that lend themselves to more inclusive, relational, and community-oriented sustainability discourses. Protestantism, by contrast, promotes narratives that emphasise measurability, performance, and individual responsibility, aligning more closely with technocratic and efficiency-based forms of sustainability communication (Koehrsen, 2015).

Such distinctions are particularly relevant in the analysis of corporate reporting, where cultural expectations shape both the content and the communicative form of disclosures. While religious affiliation is rarely made explicit in sustainability reports, its influence persists in the background, subtly guiding what is selected, emphasised, or omitted in organisational narratives.

2.2. Sociology of worth through Catholic and Protestant ethics

The Sociology of Worth (SOW), developed by Boltanski and Thévenot (1991), provides a normative framework through which social actions can be evaluated and legitimised. Subsequent organisational studies have adopted the SOW perspective to examine justification and legitimacy in complex institutional settings (Cloutier and Langley, 2013; Jagd, 2011; Patriotta et al., 2011), showing its suitability for analysing discursive practices and accountability narratives. At its core, the theory identifies multiple "orders of worth" or "worlds of value", each grounded in a distinct moral logic and centred on a specific principle that defines what is considered valuable, legitimate, or worthy of recognition in each context. These worlds reflect idealised visions of how society ought to be organised, and they offer actors a repertoire of justification strategies for interpreting and defending their actions.

In their foundational work, Boltanski and Thévenot outline seven principal worlds: market, industrial, civic, fame, domestic, inspired, and, in later extensions, green. For this study, five of these worlds have been retained: market, industrial, civic, domestic, and green. The fame and inspired worlds were excluded due to their partial conceptual overlap with others more relevant to the public utility sector. Specifically, the fame world, which is focused on visibility, reputation, and public recognition, is often embedded within the civic world, encompassing transparency and accountability in service to the common good. The inspired world, oriented around creativity and emotion, is reflected in the domestic world, where notions of commitment, loyalty, and moral identity are more structurally resonant with the communicative practices of public service organisations.

Each of the five selected worlds of worth can be meaningfully connected to the ethical orientations found in Catholic and Protestant traditions. While the SOW framework is secular in origin, its underlying logics of justification resonate with broader moral systems, including those rooted in religious worldviews. Table 1 summarises these associations and the rationale for their inclusion in the analysis of sustainability narratives within the water utility sector, building on Boltanski and Thévenot's (1991) original formulation and its subsequent organisational extensions (Cloutier and Langley, 2013; Jagd, 2011).

The market world, which centres on competition, pricing, and contractual freedom, aligns closely with Protestant ethics. Protestant traditions emphasise individual responsibility, economic autonomy, and market rationality, viewing the pursuit of profit, when coupled with discipline and restraint, as morally virtuous (Tawney, 1926). By contrast, Catholic ethics tend to assign lesser normative weight to the market logic, subordinating economic efficiency to principles of distributive justice, solidarity, and the protection of the vulnerable (Pontifical Council for Justice and Peace, 2004).

The industrial world, grounded in efficiency, performance, and technological progress, resonates with both traditions, albeit with differing emphases. Protestant ethics view industrial rationality as an expression of long-term planning and disciplined work ethic, consistent

with Weber's thesis on the Protestant spirit and capitalism (McCleary, 2007; Weber, 2003). Catholic ethics, while equally endorsing technological advancement, frame it within a broader concern for the common good, advocating for sustainable resource use and inclusive development (Francis, 2015).

The civic world prioritises equality, democratic participation, and collective welfare. It finds strong alignment with Catholic ethics, which emphasise systemic solidarity, institutional responsibility, and the moral imperative to care for the common good (Christie et al., 2019; Hollenbach, 2002). Protestant ethics also acknowledge the importance of social responsibility, though often filtered through a more individualised lens, focusing on personal initiative and localised forms of mutual aid rather than institutional redistribution (Putnam, 2000; Rowe, 2017).

The domestic world is characterised by hierarchy, trust, tradition, and moral authority. It reflects the Catholic emphasis on institutional mediation, particularly the role of the Church and family structures, in cultivating ethical behaviour. Catholic social teaching recognises the importance of continuity, interdependence, and embeddedness in moral life (Massaro, 2023). Protestant ethics, in contrast, typically prioritise individual autonomy and direct personal responsibility, thereby diminishing the prominence of the domestic world in favour of more decentralised, egalitarian forms of moral organization (Hunter, 1987; Weber, 2003).

Finally, the green world, which centres on environmental stewardship, intergenerational justice, and the intrinsic value of nature, aligns particularly well with Catholic ethics. Encyclicals such as *Laudato Si'* articulate a holistic ecological vision, calling for integral human development and environmental protection as inseparable dimensions of the common good (Francis, 2015). Protestant ethics also support environmental responsibility, though typically framed within a stewardship model, where nature is entrusted to individuals as a resource to be managed efficiently and responsibly in service of long-term societal well-being (Sherkat and Ellison, 2007).

Table 1 synthesises the conceptual logic underpinning the framework developed in this study. Each world of worth, as theorised by Boltanski and Thévenot (1991), is first defined through its core justificatory principles. These principles are then aligned with the moral orientations traditionally associated with Catholic and Protestant ethics, drawing on authoritative sources within each tradition. The resulting structure reflects a two-step reasoning process.

First, the key characteristics of each world provide a stable normative anchor. Second, these characteristics are mapped onto the distinct ethical emphases of the two religious traditions. Catholic social thought tends to frame justification in terms of solidarity, institutional responsibility, and the common good, whereas Protestant ethics often privilege autonomy, discipline, and individual responsibility.

By bringing these elements together, the table articulates the conceptual alignment that supports the framework. Each value world is interpreted through the dual lenses of its normative foundations and the ethical orientations of the two traditions, offering a coherent structure for analysing sustainability narratives. In doing so, the study extends the analytical reach of the Sociology of Worth by applying it to forms of corporate communication shaped by long-standing religious and cultural traditions.

2.3. Worlds of value in water utility reporting

The Market World frames water as an economic asset, focusing on cost-efficiency, profitability, and competitive positioning. Within this logic, sustainability is interpreted through the lens of economic performance, as reflected in recurring themes such as efficiency and productivity in water management (Bel and Warner, 2008; Walter et al., 2009). The emphasis lies on reducing operational costs and optimizing resource allocation to enhance financial returns (Bhattacharyya and Agbola, 2018; Correia and Marques, 2011; Cunha Marques, 2008; Fabbri and Fraquelli, 2000; Garde Sánchez et al., 2017; Romano and Guerrini,

Table 1
Theoretical framework.

Worlds of Worth	Key Characteristics (Boltanski and Thévenot, 1991)	Catholic Ethics	Protestant Ethics
Market	Competition, exchange, merit-based value	Subordinated to solidarity and distributive justice (Pontifical Council for Justice and Peace, 2004).	Linked to autonomy, efficiency, and moral self-reliance (Tawney, 1926).
Industrial	Efficiency, productivity, planning, innovation	Oriented toward sustainable development and the common good (Francis, 2015).	Reflects discipline, rational planning, and work ethic (McCleary, 2007; Weber, 2003).
Civic	Equality, public interest, collective welfare	Aligned with systemic solidarity and institutional responsibility (Christie et al., 2019; Hollenbach, 2002).	Related to individual contribution to social order (Putnam, 2000; Rowe, 2007).
Domestic	Hierarchy, tradition, loyalty, trust	Central to relational ethics and institutional mediation (Massaro, 2023).	Less emphasis; prioritises autonomy over hierarchy (Hunter, 1987; Weber, 2003).
Green	Environmental stewardship, intergenerational justice	Integral to the common good (Francis, 2015)	Framed as responsible resource management and duty (Sherkat and Ellison, 2007).

2011). Investments in operational improvements, particularly those that reduce water losses, serve a dual purpose: they support environmental sustainability and reshape cost structures, thereby influencing tariff levels and economic sustainability (D'Amore et al., 2024). In this context, financial performance indicators are central, linking environmental efforts to measurable economic outcomes and highlighting the role of efficiency in driving overall business performance (Alshehhi et al., 2018; Remo-Diez et al., 2023). In increasingly competitive utility markets, firms seek to differentiate themselves by integrating sustainability into strategic planning, thereby securing market share and long-term economic resilience (Bakker, 2014; D'Amore et al., 2021; O'Callaghan et al., 2020). Such integration positions sustainability not merely as a compliance requirement, but as a lever for competitive advantage in a sector where financial and environmental goals are increasingly interdependent (Mio et al., 2023; Tonnamello et al., 2025; Veltri et al., 2023).

The Industrial World is characterised by technological modernisation, infrastructure investment, and process optimisation. It reflects a logic of rationality and control, where the pursuit of efficiency underpins both service quality and environmental objectives. Within this world, companies focus on streamlining operations, reducing waste, and maximising performance through automation and standardised procedures (Cherchi et al., 2015; Lombardi et al., 2019). Technological innovation plays a pivotal role in this domain. The adoption of digital monitoring tools (Beal and Flynn, 2015) and AI-based analytics (D'Amore et al., 2022; Krishnan et al., 2022; Nti et al., 2022) enhances real-time decision-making and operational responsiveness. At the same time, infrastructure development remains a cornerstone of industrial investment, as firms modernise pipelines, expand treatment capacity, and reinforce distribution systems to improve long-term sustainability and service resilience (Furlong et al., 2019; Liu and Jensen, 2018; Taiwo et al., 2023). In this world, sustainability is embedded in the pursuit of technological excellence, where innovation and efficiency are not ends in themselves, but means to ensure reliable, cost-effective, and sustainable water services.

The Civic World interprets water as a fundamental right and a collective good, anchoring sustainability in principles of equity, transparency, and democratic participation. In this framework, water governance is conceived as a shared responsibility among institutions, civil society, and citizens (Agovino et al., 2021; Akhmouch and Correia, 2016; Romano and Akhmouch, 2019). This perspective highlights water crises as governance failures rather than technical deficiencies, calling for inclusive and accountable management approaches (Bijani et al., 2020; Casadevall, 2016; Rogers and Hall, 2003). A central element of this logic is tariff regulation, which functions as a social tool to balance affordability with financial viability (Barbosa and Brusca, 2015; Guerini et al., 2018; Pérez-Urdiales et al., 2024; Pinto and Marques, 2015). Fair pricing mechanisms help mitigate economic disparities and uphold the principle of universal access to water. Stakeholder and community engagement are also integral, promoting co-responsibility and participatory decision-making among public authorities, service providers, and civil society actors (Di Vaio et al., 2021; Jiménez et al., 2020; Tommasetti et al., 2020). This engagement fosters legitimacy and ensures that water governance reflects diverse societal needs. Public health is another key concern within the Civic World, which stresses the importance of safe drinking water, sanitation, and preventive action against waterborne diseases (Gunnarsdottir et al., 2012; Hruđey et al., 2006; Roeger and Tavares, 2018). Companies operating in this register assume a role as stewards of public welfare, integrating sustainability into broader commitments to social responsibility and human rights (Contrafatto, 2014; D'Amore et al., 2024).

The Domestic World frames water through trust, continuity, and local rootedness. Legitimacy is grounded in long-term relationships with communities, stable governance, and institutional familiarity. Utilities aligned with this world often collaborate with municipalities, regulators, and local organisations. These partnerships support governance

models tailored to local needs and long-term goals (Lima et al., 2021; Marin, 2009; Silvestre et al., 2018). The emphasis is on relational legitimacy. Transparency and social responsibility help reinforce public trust and institutional credibility (Weisner et al., 2020). Research in this area underscores the centrality of trust in shaping public attitudes, especially in sectors involving health, safety, and service reliability. It strongly influences how people perceive risk, particularly when services are managed by technical experts (Earle and Cvetkovich, 2013; Freudenburg, 1993). Clear and consistent communication builds confidence in water utilities. Research on drinking water perception and institutional trust shows that disclosure practices and reputation play a key role (Hu et al., 2011; Levallois et al., 1999). Weisner et al. (2020) stress that regular, transparent engagement is essential to maintaining trust, especially when service reliability is under public scrutiny. In this world, sustainability reporting serves a relational purpose. It is not just a compliance tool, but a way to affirm accountability and closeness to the community.

The Green World is centred on environmental stewardship, ecological responsibility, and long-term sustainability. In the water sector, this logic takes shape through ecosystem-based approaches that aim to reduce the environmental impact of infrastructure and human activity. A key concern is the protection of biodiversity and aquatic ecosystems. Many companies incorporate habitat conservation and river basin health into their strategic plans (Venturelli et al., 2023). Nature-based solutions are widely adopted. These include rainwater harvesting (Kim et al., 2025; Nandi and Gonela, 2022), green water treatment systems (Ramírez-Agudelo et al., 2021; Tsatsou et al., 2023), and climate-resilient infrastructure (Westling et al., 2019). Such initiatives aim to combine environmental integrity with operational efficiency. This environmental orientation is reinforced through adherence to global sustainability frameworks. Companies align their strategies with the UN Global Compact, the Paris Agreement, the European Green Deal, and the Sustainable Development Goals (Di Vaio et al., 2021; Manes-Rossi and Nicolo', 2022). These frameworks guide long-term planning and reflect commitment to climate goals. Transparency is ensured through reporting standards such as GRI, SASB, ESRS, and TCFD, supporting the disclosure of environmental performance and strengthen public and investor trust (D'Amore et al., 2024; De Villiers and Van Staden, 2011; Ligorio et al., 2025; Maroun and Ecim, 2024). In this world, sustainability is not just a regulatory requirement. It is a strategic and moral responsibility. Water utilities are called to act as environmental stewards, contributing to both local ecosystem resilience and global climate stability.

Table 2 summarises the key themes associated with each world of worth, offering a structured synthesis of the issues that emerge from the

Table 2
Key themes of sustainability reporting across worlds of worth.

Worlds of Worth	General Themes
Market	- Economic efficiency and productivity in water management - Key Financial Performance Indicators related to water resource management
Industrial	- Market positioning - Operational efficiency and resource optimisation - Implementation in innovative technology
Civic	- Investment in water infrastructure - Equitable access to water resources - Community and stakeholder engagement
Domestic	- Public health and water safety - Protection of water resources - Partnership with local institutions
Green	- Transparency and trust - Biodiversity conservation and protection of ecosystems - Sustainable water resource management and green infrastructure - Commitment to global climate initiatives

Source: authors' elaboration.

literature just analysed. These thematic clusters are consistent with prior analyses of sustainability reporting in the water utility sector, which document the coexistence of economic, operational, civic, relational and environmental concerns in corporate narratives (Bakker, 2014; Di Vaio et al., 2021; D'Amore et al., 2024; Manes-Rossi and Nicolo', 2022; Venturelli et al., 2023). The themes reflect recurrent patterns identified in prior studies on water utility sustainability reporting and provide a structured basis for linking these concerns to the analytical lens offered by the Sociology of Worth, informing the coding scheme used in the empirical analysis.

3. Research methodology

3.1. Research design

Following the theoretical insights developed in the previous sections, on religiosity as a source of ethical orientation, the interpretive potential of the Sociology of Worth, and the articulation of value in water utility narratives, this study adopts qualitative content analysis as its methodological approach. Content analysis is defined as “a research technique for making replicable and valid inferences from data according to their context” (Krippendorff, 1980, p.21) and is widely used in social and environmental accounting to examine non-financial disclosures. Its capacity to identify and interpret meaning patterns makes it particularly well suited to explore the ethical frameworks, justificatory logics, and value orientations embedded in corporate sustainability reporting (Manes-Rossi and Nicolo', 2022; Michelon and Parbonetti, 2012; Talbot and Boiral, 2018; Yongvanich and Guthrie, 2007).

To guide the analysis, a coding framework was developed based through a deductive operationalisation of the theoretical constructs. This strategy is essential for ensuring a structured, methodologically sound, and analytically transparent content analysis, relying on clearly defined categories derived from well-established scholarly frameworks (Harper and Thompson, 2011; Silva, 2021). This framework integrates insights from two main bodies of scholarship. The first concerns empirical studies on sustainability reporting in the water utility sector, which identify recurring thematic areas such as environmental protection, equitable access, infrastructure investment, stakeholder engagement, and service quality. These sector-specific themes were then situated within the conceptual architecture of the Sociology of Worth (Boltanski and Thévenot, 1991; Boltanski et al., 2006), which conceptualises justification through multiple “worlds of value”, each grounded in distinct normative principles. Building on the theoretical foundations outlined in the previous section, this model was interpreted through the ethical lenses of Catholic and Protestant traditions. As discussed earlier, these traditions articulate different moral understandings of responsibility, legitimacy and the common good. The coding framework was therefore constructed by deductively linking sectoral themes to specific worlds of value and interpreting these worlds considering the ethical distinctions embedded in the two religious heritages. This approach allowed for the development of a matrix capable of capturing both the explicit discursive content of sustainability reporting and the deeper moral orientations underpinning it (Patriotta et al., 2011).

The empirical analysis draws on sustainability reports published by water utilities in Italy and the United Kingdom, two national contexts where Catholic and Protestant traditions have historically contributed to shaping cultural and institutional values. The sample includes twenty-four utilities (twelve per country) operating in regulated water sectors and providing comparable infrastructural and public service functions. The organisational structures of these operators differ across the two countries. UK utilities are predominantly privately owned, whereas Italian operators are largely publicly controlled or embedded in municipal governance arrangements. These contextual characteristics are described in greater detail in the sections that follow. The study considers the most recent sustainability reports available at the time of data collection, namely the 2023 reporting cycle for the Italian utilities

and the 2024 cycle for the UK utilities. This choice reflects the objective of examining the most up-to-date and mature reporting practices in each context rather than tracing narrative developments over time. Using the coding matrix, each report was examined to assess the presence and prominence of the various worlds of value. An initial set of keywords was identified for each world of worth. These keywords were drawn from the literature on sustainability reporting, the conceptual definitions associated with each world and recurring terminology in the water utility sector. Because the corpus included documents written in both Italian and English, separate keyword lists were developed for each linguistic context. The terms were translated and cross-checked to ensure conceptual and semantic equivalence across the two languages. Attention was paid to maintaining alignment between keywords and the justificatory meanings associated with each world of worth.

Pearson's Chi-square test was applied to evaluate the discriminative power of each candidate keyword across the corpus. Following best practices in feature selection (Forman, 2003), only terms with the strongest discriminative capacity were retained. A high cut-off threshold ($\chi^2 > 100$) was adopted, consistent with evidence showing that high Chi-square values identify terms that meaningfully differentiate documents, while lower thresholds tend to retain noisy or weakly informative features (Zheng et al., 2004). This procedure ensured that only the most contextually salient and analytically robust keywords were included in the final matrix.

The refined keyword sets were used to analyse sustainability reports from Italian and UK water utilities, two contexts shaped historically by Catholic and Protestant traditions, respectively. Using MAXQDA, a software widely used in qualitative academic research (Kuckartz and Rädiker, 2019), the frequency of occurrence of each keyword was calculated. These frequencies provided an initial metric for assessing the presence and prominence of each world of value within the narratives (Krippendorff, 1980; Neuendorf, 2017). The analysis focused on identifying the themes addressed, the justificatory strategies employed, and the forms of legitimacy constructed, with particular attention to cross-country differences in the value narratives structuring sustainability reporting. However, it is important to clarify that the purpose of the analysis is not to statistically compare Italy and the United Kingdom on the basis of absolute keyword counts. The sustainability reports produced in the two countries differ substantially in structure, length and narrative density, with Italian reports generally being more extensive and detailed, and raw frequencies therefore do not constitute equivalent units of analysis. For this reason, the evaluation focuses on the internal distribution of worlds of worth within each national context, assessing the relative prominence of each world rather than conducting cross-country inferential tests on non-comparable textual corpora.

In total, 12 keywords were selected for the Market World, 9 for the Industrial World, 11 for the Civic World, 8 for the Domestic World, and 13 for the Green World. This structured keyword-based approach enabled a comparative and quantifiable assessment of the prominence of each value world within the two national contexts, providing a solid empirical foundation for interpreting the ethical and cultural underpinnings of sustainability reporting practices (Fig. 1).

Finally, the last step involved assigning a qualitative weight to each world based on its relative frequency within the reports (see Table 3). The proportion of keyword occurrences associated with each world was calculated against the total number of references in each national context. Worlds were categorised as high (>20% of references), medium (10–20%) or low (<10%). These thresholds are used as heuristic cut-off points (Neuendorf, 2017), in line with common practice in content analysis and empirical applications of the Sociology of Worth and sustainability reporting studies (see Boiral, 2013; Patriotta et al., 2011). Applying them separately to the Italian and UK corpora ensured sensitivity to contextual variation and avoided the methodological fallacy of assuming symmetrical relevance of value worlds across cultural settings. This rule-based weighting procedure reduced subjective interpretation and enhanced internal consistency.

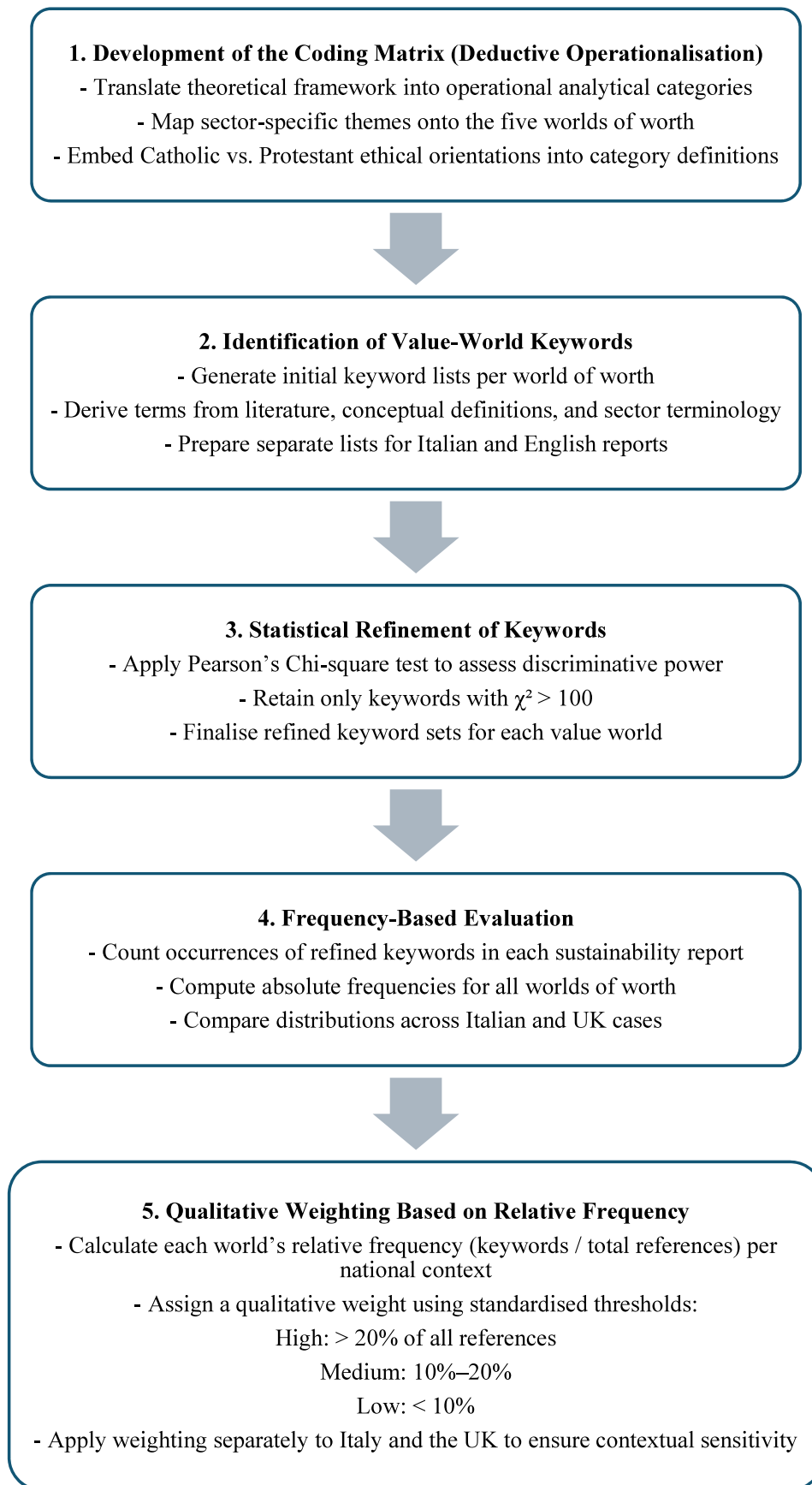


Fig. 1. Flow Chart of Methodology. Source: authors' elaboration.

Table 3
Coding framework.

Worlds of Value	General Themes	Italian Words	English Words	Frequency	
Market	Economic efficiency and productivity in water management Key Financial Performance Indicators related to water resource management Market positioning	efficienza	efficiency	596	209
		performance	performance	584	985
		costo	cost	152	220
		utile	profit	98	87
		rendimento	return	83	66
		ebitda	ebitda	73	14
		concorrenza	competitor	72	16
		azionista	shareholder	22	101
		rischio	risk	1027	1342
		cliente	customer	385	480
		reddito	income	43	103
		bond	bond	98	181
		innovazione	innovation	692	127
		digitale	digital	416	43
Industrial	Operational efficiency and resource optimisation Implementation in innovative technology Investment in water infrastructure	investimento	investment	169	563
		tecnologia	technology	157	180
		pianificazione	plan	155	703
		ottimizzazione	optimisation	112	9
		infrastruttura	infrastructure	101	251
		operativo	operational	155	389
		processo	process	927	267
		sicurezza	safety	1520	170
		stakeholder	stakeholder	1279	122
		salute	health	861	287
Civic	Equitable access to water resources Community and stakeholder engagement Public health and water safety	materialità	materiality	403	72
		engagement	engagement	680	272
		inclusione	inclusion	307	92
		partecipazione	participation	261	7
		equità	fairness	82	193
		tariffa	tariff	61	39
		vulnerabilità	vulnerability	32	157
		sociale	social	758	321
		relazione	relation	783	120
		collaborazione	collaboration	410	74
		trasparenza	transparency	257	66
		educazione	education	92	32
Domestic	Protection of water resources Partnership with local institutions Transparency and trust	locale	local	150	164
		fiducia	trust	56	100
		partenariato	partnership	11	120
		comunità	community	778	215
		ambiente	environment	1671	1155
		impatto	impact	830	570
		biodiversità	biodiversity	546	355
		naturale	nature	528	547
		green	green	296	175
		clima	climate	716	849
Green	Biodiversity conservation and protection of ecosystems Sustainable water resource management and green infrastructure Commitment to global climate initiatives	emissione	emission	142	28
		carbon	carbon	60	672
		ecosistema	ecosystem	128	47
		rifiuto	waste	121	158
		sdgs	sdgs	129	50
		circolare	circular	608	29
		riciclo	recycle	118	132

Source: authors' elaboration.

To increase the interpretive robustness of the keyword-based coding procedure, a small set of representative excerpts from the reports was manually reviewed. The purpose of this step was not to validate the frequency results, which are derived from the systematic keyword analysis, but to verify that the terms included in the dictionary were used in contexts consistent with the justificatory logics associated with each world of worth. This check was intended to ensure that the keywords captured meaningful narrative justifications rather than isolated lexical occurrences. Table 4 presents illustrative excerpts from both the UK and Italian reports that exemplify how the different worlds of worth are articulated within the sustainability narratives.

3.2. Empirical context and case selection

Before examining the two national samples in detail, it is necessary to clarify the grounds for their comparability and analytical relevance. As

shown in Tables 5 and 6, the Italian and UK water sectors differ in ownership structures, governance models, and reporting traditions. However, they also share several fundamental characteristics that justify a comparative analysis. In both countries, water utilities operate as public-interest entities that deliver an essential service under a defined regulatory framework. They are also increasingly expected to provide structured sustainability disclosures. These similarities ensure that sustainability reporting performs comparable functions in both contexts. It is used to communicate performance, accountability, and long-term commitments, even when organisational forms differ.

At the same time, the two contexts present meaningful contrasts. The UK sector is predominantly privately owned and shows fragmented and heterogeneous reporting practices. Italian utilities, by contrast, are largely publicly controlled and display a higher degree of standardisation in sustainability communication. These institutional differences coexist with broader ethical and cultural traditions historically

Table 4
Illustrative excerpts associated with the worlds of worth.

Words of Value	UK report excerpt	Italy report excerpt
Market	“There are further opportunities to advance our use of data intelligently, alongside energy efficiency projects aimed at optimizing energy use on site.” “The project benefited from the use of BIM 3D modelling, which enabled improved project and asset information and better management of associated costs, delivery timeline and risk.”	“The implementation of smart sensors and data analysis technologies could increase the company's efficiency and reduce operating costs.” [Original version: <i>“L'implementazione di sensori intelligenti e di tecnologie di analisi dei dati potrebbe aumentare l'efficienza dell'azienda e ridurre i costi operativi.”</i>] “Greater commitment is required in the activities of planning purchases and procurement procedures, although actions aimed at improving the timeliness, effectiveness and efficiency of the process have been implemented.” [Original version: <i>“Un maggiore impegno è richiesto nelle attività di programmazione degli acquisti e degli affidamenti, benché siano state implementate azioni tese a migliorare tempestività, efficacia ed efficienza del processo.”</i>]
Industrial	“Investment in infrastructure, such as our West-East Link Main and West Cumbria pipeline, allows us to transfer water around the region more efficiently to avoid depletion of individual water sources.” “We plan far into the future and invest in our infrastructure to ensure sustainability and operational resilience.”	“The industrial research and experimental development project co-financed by MIUR addressed, through an innovative multi-objective approach, the issues of managerial and functional rationalisation, efficiency improvement and energy optimisation of infrastructural systems.” [Original version: <i>“Il progetto di ricerca industriale e sviluppo sperimentale cofinanziato dal MIUR, ha affrontato mediante un approccio multi-obiettivo innovativo i temi della razionalizzazione gestionale e funzionale, dell'efficientamento e dell'ottimizzazione energetica dei sistemi infrastrutturali.”</i>] “To this end, the Industrial Plan provides for a ‘cloud project’ for the migration of the Group's entire IT infrastructure to the cloud, to improve security, performance and scalability.” [Original version: <i>“A tal fine, il Piano industriale prevede un ‘progetto cloud’ per la migrazione sul cloud dell'intera infrastruttura IT del Gruppo, con l'obiettivo di migliorare la sicurezza, le performance e la scalabilità.”</i>]
Civic	“We act with integrity, giving suppliers confidence in the way we do business, which translates to transparency and fairness for everyone that works with us.” “Ensuring that we have a safe, healthy, happy, and skilled workforce is essential for us to provide our services.”	“We are committed to achieving pay equity, equality in personnel selection and organisational processes, equal opportunities for career advancement, prevention of all forms of physical, verbal and digital abuse in the workplace, work–life balance, and the protection of parenthood and care work.” [Original version: <i>“Ci impegniamo a raggiungere equità delle retribuzioni, parità nei processi di selezione e organizzazione del personale, pari possibilità di crescita di carriera, prevenzione di ogni forma di abuso fisico, verbale e digitale sul luogo di lavoro,</i>

Table 4 (continued)

Words of Value	UK report excerpt	Italy report excerpt
Domestic	“Ensuring strong levels of trust is a priority for us at a time when public confidence in the water sector has declined in recent years.” “Over the past two years we've built partnerships with specialist referral partners, invested in outreach to under-represented community groups, removed barriers for the most underrepresented in society and introduced employability programmes that support those who need it most.”	<i>conciliazione dei tempi vita lavoro, tutela della genitorialità e del lavoro di cura.”</i> “The interventions implemented produce positive effects on the territory in terms of employment and public health.” [Original version: <i>“Gli interventi realizzati producono delle ricadute positive sul territorio in termini occupazionali e di salute pubblica.”</i>] “Finally, A2A is actively committed to the efficiency improvement and safety enhancement of the existing facilities in the territories in which it operates, constantly monitoring its activities.” [Original version: <i>“Infine, A2A si impegna attivamente nell'efficientamento e messa in sicurezza delle strutture esistenti nei territori in cui opera, monitorando costantemente le proprie attività.”</i>] “AQP provides essential services to the community, for which reason the company is constantly committed to ensuring a working environment that promotes inclusiveness and motivation, to best express the potential of each individual and improve the dynamics of communication and listening within the organisation.” [Original version: <i>“AQP fornisce servizi essenziali alla comunità, motivo per cui l'azienda si impegna costantemente ad assicurare un ambiente di lavoro che favorisca l'inclusività e a motivazione, al fine di esprimere al meglio il potenziale di ogni individuo e migliorare le dinamiche di comunicazione e ascolto all'interno dell'organizzazione.”</i>]
Green	“As a public body in Scotland, we reflect the duty to “further the conservation of biodiversity” in the way we deliver our core services – managing water resources and recycling Scotland's waste waters in ways that support good ecological status.” “We are conscious of the weight of responsibility we bear to deliver safe, clean water and recycle it effectively and to protect and enhance our environment.”	“We are committed to improving the chemical and physical parameters of the water released at the end of the treatment process, complying with regulations and safeguarding biodiversity.” [Original version: <i>“Ci impegniamo a migliorare i parametri chimici e fisici dell'acqua rilasciata al termine del processo di depurazione, rispettando la normativa e salvaguardando la biodiversità.”</i>] “We have experimented with innovative technologies which, from a circular economy and energy sustainability perspective, have enabled the valorisation of sewage sludge from civil wastewater through the extraction/production of biofuels.” [Original version: <i>“Abbiamo sperimentato tecnologie innovative, che in un'ottica di economia circolare e di sostenibilità energetica, hanno consentito la valorizzazione dei fanghi di depurazione di reflui civili attraverso l'estrazione/produzione di biocarburanti.”</i>]

Table 5
Overview of UK water utilities.

Company	Ownership	Listed/Not listed	Type of Report	Multiutility
Thames Water	Private	Not listed	Sustainability Report	No
United Utilities	Public	Listed	Sustainability Report	No
Anglian Water	Private	Not listed	Integrated Report	No
Severn Trent Water	Public	Listed	Sustainability Report	No
Scottish Water	Public	Not listed	Biodiversity Report	No
Yorkshire Water	Private	Not listed	Impact Report	No
Southern Water	Private	Not listed	Annual Report	No
Wessex Water	Private	Not listed	Impact Report	No
South East Water	Private	Not listed	Annual Report	No
Pennon	Public	Listed	Sustainability Report	No
South Staffs Water	Private	Not listed	Sustainability Report	No
Welsh Water	Not-for-Profit	Not listed	ESG Strategy	No

Source: authors' elaboration.

Table 6
Overview of Italian water utilities.

Company	Ownership	Listed/Not listed	Type of Report	Multiutility
Hera	Mixed	Yes	Sustainability Report	Yes
A2A	Mixed	Yes	Integrated Report	Yes
Iren	Mixed	Yes	Sustainability Report	Yes
Acea	Mixed	Yes	Sustainability Report	Yes
Dolomiti Energia	Mixed	Yes	Non-Financial Statement	Yes
Acinque	Public	Yes	Sustainability Report	Yes
SMAT	Public	No	Sustainability Report	No
AqP	Public	No	Integrated Report	No
Gruppo CAP	Public	No	Sustainability Report	No
Aimag	Mixed	No	Sustainability Report	Yes
MM	Public	Yes	Sustainability Report	No
Gori	Mixed	No	Sustainability Report	No

Source: authors' elaboration.

associated with Protestant and Catholic contexts. Such traditions have contributed to shaping societal ideas of responsibility, legitimacy, and collective welfare. These traditions may also provide broader ethical and cultural repertoires that shape how responsibility, legitimacy, and collective welfare are interpreted within society. In this sense, they can be seen as background moral repertoires that may interact with governance arrangements, ownership structures, and regulatory incentives. The Italian and UK cases, therefore, combine structural similarity with normative divergence. This makes them a theoretically robust pair for comparison. It allows the analysis to explore how ethical and cultural repertoires intersect with institutional design in shaping sustainability narratives, while maintaining comparable sectoral functions and regulatory expectations across the two contexts.

3.2.1. UK context

The UK sample comprises twelve water utilities selected to reflect the diversity of organisational forms and reporting practices within the sector. A clear majority of the companies analysed are privately owned, with only three publicly listed firms (United Utilities, Severn Trent Water, and Pennon) and one non-profit entity (Welsh Water). No multi-utility companies were included, ensuring a sector-specific focus and facilitating comparability across cases.

One notable feature of the UK context is the heterogeneity in the modes of sustainability communication. Companies adopt a variety of reporting formats, including Sustainability Reports, Integrated Reports, Impact Reports, Annual Reports, ESG strategies, and even Biodiversity Reports. This variation suggests that, despite operating in a common regulatory and sectoral environment, organisations differ significantly in how they articulate their sustainability performance and define relevant themes and audiences.

The sample thus reflects a predominantly private and structurally fragmented sector, characterised by a diversity of disclosure approaches and no centralised model of reporting.

3.2.2. Italy context

The Italian sample includes twelve water utilities, selected to represent the organisational and reporting diversity within the national context. Unlike the UK case, most companies analysed are publicly owned or public-private partnerships. Only three companies - Hera, A2A, and Iren - are publicly listed multi-utilities with a significant degree of private shareholding. However, both Hera and A2A remain under public control due to substantial municipal ownership and shareholder agreements, while Iren is predominantly privately owned. This reflects the structural distinction in Italy between large, market-oriented utilities and territorially embedded public operators.

A second key feature of the Italian context is the high level of standardisation in sustainability communication. Almost all companies publish a standalone Sustainability Report, with only two exceptions (A2A and AQP), which adopt an Integrated Report format, and Dolomiti Energia, which discloses through a Non-Financial Statement. This consistency suggests a widespread institutionalisation of sustainability reporting practices, likely driven by national regulatory requirements and sectoral convergence.

Furthermore, the presence of multi-utility firms, especially among listed companies, highlights a dual logic: larger players tend to integrate water services into broader corporate structures, while smaller public operators retain a more focused identity. This structural difference may influence the articulation of value and the framing of sustainability.

4. Findings

The distribution of value worlds across UK and Italian water utility reports reveals both convergences and notable divergences in the narratives employed (see Table 7).

The Green World emerges as the most prominent in both contexts - high in the UK (0,35) and Italy (0,28) - indicating a shared prioritisation of environmental issues and alignment with global sustainability agendas. This convergence suggests that ecological concerns, such as climate change and biodiversity protection, represent a transnational

Table 7
Salience levels of value worlds across the UK and Italy.

World of Value	UK		Italy	
Market	0,28	HIGH	0,16	MEDIUM
Industrial	0,18	MEDIUM	0,14	MEDIUM
Civic	0,13	MEDIUM	0,30	HIGH
Domestic	0,06	LOW	0,12	MEDIUM
Green	0,35	HIGH	0,28	HIGH

Source: authors' elaboration.

imperative that transcends national boundaries and institutional models. However, differences emerge in the relative emphasis placed on other worlds.

The Market World holds high intensity in the UK (0,28) but only medium in Italy (0,16), reflecting the stronger presence of market-oriented narratives in the UK's privatised and investor-driven water sector, where risk, performance, and financial value are central to legitimacy. In contrast, Italian utilities, often publicly owned, articulate market logics more moderately, balancing them with social and public-oriented discourses.

The Civic World shows the reverse pattern: it is high in Italy (0,30) but only medium in the UK (0,13). This suggests that Italian utilities place greater discursive weight on social inclusion, public health, and stakeholder engagement, consistent with their public service mandate and embeddedness in local communities. UK reports, while acknowledging community engagement, tend to frame these efforts more in terms of customer service and procedural fairness than deep social commitments.

The Domestic World appears medium in Italy (0,12) but low in the UK (0,06), pointing to a stronger Italian emphasis on trust, relational embeddedness, and institutional continuity - elements that are less central in the UK context, where legitimacy is more often derived from technical performance and transparency.

Lastly, the Industrial World is of medium intensity in both contexts (0,18 UK - 0,14 Italy), although the UK utilities tend to articulate it through highly structured and data-driven narratives tied to planning, innovation, and efficiency, whereas Italian utilities often present such themes with more heterogeneity, especially among smaller, publicly owned firms. Overall, the findings reflect how different governance models and institutional traditions shape the salience of value worlds, with the Green World acting as a unifying discourse, while others vary according to national priorities and organisational structures.

4.1. Market world

Table 8 highlights notable divergences in how the Market World is constructed in the sustainability narratives of Italian and UK water utilities, reflecting distinct institutional logics and ownership structures. In the Italian context, the dominant presence of terms such as “rischio” (risk), “efficienza” (efficiency), and “performance” underscores a market discourse primarily concerned with institutional resilience, regulatory compliance, and cost-effectiveness. These patterns align with the public or publicly controlled nature of most Italian utilities, where market legitimacy is pursued through operational sustainability and balanced budgeting rather than shareholder value maximization. In this light, the low incidence of the term “azionista” (shareholder) can be interpreted

not as a lack of concern for economic performance, but as a reflection of the limited role that equity holders play in governance structures, given the predominance of municipal or regional ownership.

In the UK context, the centrality of “risk” and “performance” reveals a similar concern with financial and operational accountability, yet within a different institutional environment-one marked by full privatisation and investor oversight. The relatively high frequency of “customer” indicates that service quality and user satisfaction are key elements of market-based legitimacy, particularly within a price-cap regulatory model. Interestingly, “ebitda” and “competitor” register minimal presence. The negligible role of “ebitda” suggests that, despite being profit-oriented, UK utilities prefer to frame their economic performance through outcome-based or service-level indicators rather than traditional financial metrics, possibly due to the sensitivity around profit-making in a privatised essential service. The absence of “competitor” reflects the monopolistic nature of regional water provision in the UK, where utilities operate in exclusive service areas and are benchmarked not against direct market rivals, but through comparative performance reviews overseen by Ofwat.

4.2. Industrial world

In the Industrial World, the sustainability narratives of Italian and UK water utilities reveal distinct configurations, reflecting broader structural and institutional differences (see Table 9). In the Italian context, the most frequently cited keywords - “processo” (process), “innovazione” (innovation), and “digitale” (digital) - suggest a discursive emphasis on internal organisational development, incremental innovation, and digital transformation as levers of operational improvement. These terms construct a narrative centred on ongoing optimisation of service delivery rather than large-scale infrastructural renewal. By contrast, keywords such as “infrastruttura” (infrastructure) and “investimento” (investment) occur with lower frequency, indicating that capital expenditure and long-term planning are less central in the reporting discourse. This pattern is coherent with the financial constraints and public ownership that characterise the Italian sector, where large investment projects often depend on external public funding and are framed less in terms of strategic expansion and more in aspirational or compliance-related language (D'Amore et al., 2022).

In the UK context, the narrative shifts decisively toward forward-looking planning and financial commitment, as shown by the high relative frequency of “plan”, “investment”, and “operational”. These keywords reflect a sector structured around regulatory targets and shareholder accountability, where corporate legitimacy is strongly tied to measurable performance and capital-intensive strategies. The prominence of investment-related terminology aligns with the privatised

Table 8
Findings for the market world.

Market					
Italy			UK		
Keyword	Absolute Frequency	Relative Frequency	Keyword	Absolute Frequency	Relative Frequency
efficienza	596	0,184	efficiency	209	0,055
performance	584	0,181	performance	985	0,259
costo	152	0,047	cost	220	0,058
utile	98	0,030	profit	87	0,023
rendimento	83	0,026	return	66	0,017
ebitda	73	0,023	ebitda	14	0,004
concorrenza	72	0,022	competitor	16	0,004
azionista	22	0,007	shareholder	101	0,027
rischio	1027	0,318	risk	1342	0,353
cliente	385	0,119	customer	480	0,126
reddito	43	0,013	income	103	0,027
bond	98	0,030	bond	181	0,048
TOT	3233		TOT	3804	

Source: authors' elaboration.

Table 9
Findings for the industrial world.

Industrial					
Italy			UK		
Keyword	Absolute Frequency	Relative Frequency	Keyword	Absolute Frequency	Relative Frequency
innovazione	692	0,240	innovation	127	0,050
digitale	416	0,144	digital	43	0,017
investimento	169	0,059	investment	563	0,222
tecnologia	157	0,054	technology	180	0,071
pianificazione	155	0,054	plan	703	0,278
ottimizzazione	112	0,039	optimisation	9	0,004
infrastruttura	101	0,035	infrastructure	251	0,099
operativo	155	0,054	operational	389	0,154
processo	927	0,321	process	267	0,105
TOT	2884		TOT	2532	

Source: authors' elaboration.

nature of the UK industry, in which firms are expected to justify tariff structures through ambitious infrastructure plans submitted to regulators. Meanwhile, the relatively low presence of “optimisation” and “digital” suggests that such themes, while present in practice, are not central to the discursive construction of industrial value in reporting. Rather than focusing on internal processes or cultural narratives of innovation, UK companies emphasise structured planning and delivery frameworks, portraying themselves as competent, reliable actors within a tightly regulated market.

4.3. Civic world

In the Civic World, the analysis reveals differing emphases between Italian and UK water utilities, shaped by their respective socio-institutional environments and ethical traditions (see Table 10). In the Italian reports, the most frequently occurring keywords – “stakeholder”, “sicurezza” (safety), and “salute” (health) - underscore a strong focus on physical well-being and risk prevention as central elements of civic responsibility. This emphasis likely reflects the operational priorities of water utilities, which must ensure safe and continuous service delivery in contexts often shaped by ageing infrastructure and environmental stressors. The centrality of “stakeholder” suggests a concern for engaging with a broad spectrum of actors, although the term is commonly used in formal, institutional contexts, such as stakeholder mapping or public consultations, rather than in reference to deep participatory practices. Conversely, the limited use of terms like “tariffa” (tariff) and “vulnerabilità” (vulnerability) points to a less explicit engagement with the social implications of water pricing and user hardship. These themes may be addressed through operational tools (e. g., social tariffs), but they remain marginal in the discursive frame of

civic value. This pattern suggests that while Italian utilities demonstrate awareness of civic responsibility, their communication tends to prioritise institutional performance and public health over explicit commitments to equity and inclusion.

In the UK context, civic narratives are structured differently. The most cited terms - “health”, “engagement”, and “fairness” - signal a more procedural and rights-oriented articulation of civic legitimacy. “Health”, for instance, is frequently linked to regulatory compliance and performance standards, underscoring the role of utilities as service providers accountable to oversight bodies rather than embedded in a moral framework of communal care. “Engagement” and “fairness” are central to demonstrating customer-centric governance, but are typically framed through customer satisfaction metrics, service delivery frameworks, or consultations, rather than deep participatory mechanisms. The relatively low frequencies of “tariff” and “participation” are revealing: the former reflects a normalised understanding of price-setting within regulatory regimes rather than a politically sensitive issue, while the latter suggests that while stakeholder communication is prominent, active participation is not structurally embedded in corporate discourse.

4.4. Domestic world

The analysis of the Domestic World reveals distinct narrative patterns between the Italian and UK water utility reports (see Table 11). In the Italian context, the most prominent keywords are “comunità” (community), “collaborazione” (collaboration), and “relazione” (relation), which together account for most references within this world. This lexical pattern reflects a communication strategy that emphasises proximity, co-responsibility, and a continuous, dynamic interaction with local actors. In particular, the high frequency of “relazione” is

Table 10
Findings for the civic world.

Civic					
Italy			UK		
Keyword	Absolute Frequency	Relative Frequency	Keyword	Absolute Frequency	Relative Frequency
sicurezza	1520	0,243	safety	170	0,098
stakeholder	1279	0,205	stakeholder	122	0,070
salute	861	0,138	health	287	0,166
materialità	403	0,065	materiality	72	0,042
engagement	680	0,109	engagement	272	0,157
inclusione	307	0,049	inclusion	92	0,053
partecipazione	261	0,042	participation	7	0,004
equità	82	0,013	fairness	193	0,111
tariffa	61	0,010	tariff	39	0,023
vulnerabilità	32	0,005	vulnerability	157	0,091
sociale	758	0,121	social	321	0,185
TOT	6244		TOT	1732	

Source: authors' elaboration.

Table 11
Findings for the domestic world.

Domestic					
Italy			UK		
Keyword	Absolute Frequency	Relative Frequency	Keyword	Absolute Frequency	Relative Frequency
relazione	783	0,309	relation	120	0,135
collaborazione	410	0,162	collaboration	74	0,083
trasparenza	257	0,101	transparency	66	0,074
educazione	92	0,036	education	32	0,036
locale	150	0,059	local	164	0,184
fiducia	56	0,022	trust	100	0,112
partenariato	11	0,004	partnership	120	0,135
comunità	778	0,307	community	215	0,241
TOT	2537		TOT	891	

Source: authors' elaboration.

especially significant: the term in Italian carries a broader meaning than its English counterpart relationship. It conveys not only the existence of formal or informal ties, but also the quality and continuity of those connections over time. The use of “relazione” in sustainability reporting often signals the utility's embeddedness in the social fabric of the territory and a sense of accountability that is cultivated through ongoing interaction with municipalities, schools, cultural organisations, and citizens. Rather than describing one-off initiatives or transactional interactions, “relazione” evokes a reciprocal process of mutual recognition and shared responsibility. The relatively lower frequency of “partenariato” and “fiducia” may indicate that trust and institutional cooperation are not necessarily framed as distinct or strategic objectives, but rather as outcomes embedded in the very notion of sustained “relazione”.

In the UK context, the most frequently used terms are “community”, “local”, and “partnership”, suggesting a more formalized and structured approach to local engagement. The high frequency of “partnership” indicates that UK utilities tend to frame their relationship with local stakeholders through defined roles, agreements, and collaborative programs, rather than through narrative expressions of proximity or mutual care. The recurrence of “local” further reinforces the territorial anchoring of utilities' activities, but in a functional rather than affective sense. Conversely, the low frequency of “education” and “transparency” suggests a limited focus on public awareness or dialogic interaction, possibly indicating that efforts to build legitimacy are mediated more by service quality and performance than by proactive communication or relational investment.

4.5. Green world

As shown in Table 12, in the Green World both Italian and UK utilities exhibit a high lexical engagement with environmental themes, yet the nature and orientation of this engagement differ notably across contexts. In the Italian reports, the most frequently used keywords include “impatto” (impact), “clima” (climate), “circolare” (circular), and “ambiente” (environment). This suggests a discursive focus on the systemic and long-term implications of environmental sustainability, often framed in terms of qualitative transformation and collective responsibility. The salience of “circolare” indicates a strong rhetorical and strategic emphasis on circular economy principles, possibly reflecting both EU policy agendas and local public expectations. However, the low frequency of “carbon” reveals a relatively limited engagement with carbon accounting or decarbonisation targets, pointing to a weaker alignment with internationally standardised climate reporting metrics and tools. This may be due to lower regulatory pressure, less advanced data collection infrastructure, or a broader conceptualisation of sustainability that prioritises socio-ecological regeneration over carbon quantification.

In contrast, the UK reports place strong emphasis on “carbon”, “climate”, “impact”, and “environment”, all of which rank among the most frequently used keywords. The high lexical weight of “carbon” and “climate” reflects a deeply embedded focus on climate mitigation and emissions control, aligned with the regulatory framework imposed by Ofwat and broader UK commitments to net-zero transitions. The recurrent use of terms like “impact” and “environment” further supports a performance-oriented narrative in which environmental stewardship is closely tied to risk management, measurable outcomes, and investor

Table 12
Findings for the green world.

Green					
Italy			UK		
Keyword	Absolute Frequency	Relative Frequency	Keyword	Absolute Frequency	Relative Frequency
ambiente	1671	0,284	environment	1155	0,242
impatto	830	0,141	impact	570	0,120
biodiversità	546	0,093	biodiversity	355	0,074
naturale	528	0,090	nature	547	0,115
green	296	0,050	green	175	0,037
clima	716	0,122	climate	849	0,178
emissione	142	0,024	emission	28	0,006
carbon	60	0,010	carbon	672	0,141
ecosistema	128	0,022	ecosystem	47	0,010
rifiuto	121	0,021	waste	158	0,033
sdgs	129	0,022	sdgs	50	0,010
circolare	608	0,103	circular	29	0,006
riciclo	118	0,020	recycle	132	0,028
TOT	5893		TOT	4767	

Source: authors' elaboration.

expectations. Notably, the keyword “circular” is significantly less used in the UK corpus, indicating that while circularity may be acknowledged, it does not currently serve as a dominant organizing principle in sustainability narratives. Similarly, the low presence of “sdgs” suggests that the UN’s global goals are less explicitly invoked, possibly reflecting a preference for nationally contextualized or sector-specific frameworks. Overall, the contrast between the two contexts is most evident in the inverse prominence of “carbon” and “circolare”, revealing two divergent discursive strategies: one rooted in regulatory precision and climate metrics (UK), the other in broader systemic visions of sustainability (Italy).

5. Discussion

The comparative analysis of sustainability reporting in the Italian and UK water utility sectors reveals how differing ethical-religious traditions may inform not only the content but also the structure and tone of value narratives. These narratives are not merely stylistic choices; rather, they reflect culturally embedded conceptions of responsibility, legitimacy, and institutional purpose that develop within specific institutional contexts. In both countries, economic, industrial, civic, domestic, and environmental dimensions of sustainability are mobilised through distinct discursive strategies, yet their relative emphasis varies. While these differences partly reflect variations in governance models, ownership structures, and regulatory environments, they can also be interpreted considering broader cultural traditions that shape the moral language through which organisations justify their actions.

These patterns confirm that sustainability reporting functions as a key lever of accountability for utilities (Traxler and Greiling, 2019; Imperiale et al., 2023), translating their institutional role in the provision of essential public services into culturally resonant narratives of responsibility and legitimacy.

In both Italy and the UK, economic considerations feature prominently in sustainability narratives, underscoring the pervasive influence of market logics in the sector (D'Amore et al., 2024; Mahmood and Uddin, 2021; Morris et al., 2023). However, the ethical underpinnings differ. In the UK, economic rationality is treated as self-evident and integral to corporate legitimacy, in line with a Protestant ethic that emphasises individual responsibility, performance-based accountability, and contractual reasoning (McCleary, 2007; Rowe, 2007). Financial disclosure and strategic planning are foregrounded with little need for moral elaboration. In Italy, by contrast, even publicly owned utilities adopt a market-oriented discourse, but often through a more explanatory and justificatory tone. This suggests a residual ethical framework rooted in Catholic social thought, where economic equilibrium is seen as a prerequisite for serving the common good, and where business must maintain coherence with broader societal expectations (Hollenbach, 2002; Pontifical Council for Justice and Peace, 2004).

The industrial dimension has a medium weight in both countries, though its articulation reflects contrasting institutional logics. In the UK, technological innovation and operational efficiency are framed as ends in themselves, tools for differentiation, investor confidence, and regulatory compliance (Battisti and Stoneman, 2010). The narrative is firmly anchored in a technocratic logic that aligns with Protestant ideals of rational planning and systemic discipline (Weber, 2003). In Italy, while larger utilities adopt similar narratives, smaller or publicly funded companies tend to describe innovation in aspirational terms, often linked to access to public funding rather than internal strategic direction. Here, technology is framed less as a driver of competitiveness and more as an instrument of collective benefit, reflecting a view of infrastructure as a public asset and technological advancement as a means of fostering inclusion and sustainability (Boeri et al., 2017; Cifolelli et al., 2024).

The civic and domestic dimensions exhibit the most marked divergence between the two contexts. Italian utilities, especially those under public ownership, construct a relational narrative grounded in

community, social cohesion, and participatory governance. This emphasis corresponds to a Catholic worldview in which institutions are expected to mediate between individual and collective needs and where public trust is built through proximity, inclusion, and shared ethical purpose (Arnone et al., 2024; Bartoletti and Faccioli, 2016). In the UK, by contrast, civic engagement is more formalized and often presented as a regulatory requirement or a quality assurance mechanism. The underlying logic privileges procedural transparency and service reliability, in line with a Protestant ethic that places legitimacy in individual performance rather than in collective embeddedness (McCleary, 2007). The domestic dimension, with its focus on trust, continuity, and territorial rootedness, is especially pronounced in Italy and largely absent in the UK, where privatisation and functional abstraction tend to weaken relational ties with local communities.

Environmental sustainability emerges as a shared concern across both countries, with companies in each context articulating robust narratives around climate action, resource management, and long-term resilience. Yet the framing diverges in its ethical inflexion. UK reports typically embed environmental responsibility within corporate strategy and risk governance, consistent with a view that sustainability must be demonstrably efficient, measurable, and aligned with business objectives (Friedman and Miles, 2001; Moussa and Elmarzouky, 2024). This pragmatism reflects a rational stewardship model that sees ecological commitment as a functional component of corporate performance (Baker et al., 2023; Gray, 2010). In Italy, environmental narratives are more heterogeneous. While some companies emulate the UK’s results-oriented approach, others emphasise moral responsibility, public awareness, and intergenerational justice. This dual orientation reflects a hybrid model that integrates global standards with local ethical expectations, revealing how environmental themes can both transcend and reflect cultural divides (Caputo et al., 2021). The shared prominence of environmental sustainability across both contexts also suggests a partial convergence, where the urgency of ecological challenges prompts alignment despite divergent ethical legacies.

In sum, Catholic and Protestant traditions can be understood as background moral repertoires that interact with institutional design rather than as direct determinants of organizational behaviour. These traditions provide historically rooted interpretive frames through which organisations articulate legitimacy, responsibility, and collective purpose. In the water sector, such moral repertoires intersect with sectoral histories, regulatory frameworks, and ownership configurations to produce distinct narrative configurations in sustainability reporting. Understanding this interaction is essential for interpreting corporate sustainability practices and for designing accountability frameworks that are sensitive to both institutional arrangements and underlying value systems. In this way, the analysis contributes to the emerging literature on religiosity and non-financial reporting (Farooq et al., 2019; Terzani and Turzo, 2021) by showing how religious-ethical traditions can influence the salience and articulation of value worlds within sustainability disclosure, in interaction with governance and regulatory contexts.

6. Conclusions

6.1. Theoretical Implications

This study contributes to the theorisation of sustainability reporting by advancing an ethically grounded, culturally sensitive interpretation of corporate disclosure practices. The findings challenge the assumption of universalism in sustainability reporting frameworks, demonstrating that non-financial disclosures are not culturally neutral but are instead shaped by deeper normative infrastructures—specifically, the ethical orientations historically embedded in religious traditions and broader institutional configurations. In doing so, the study responds to recent calls for more context-sensitive and ethically grounded approaches to sustainability reporting research (Brunelli and Di Carlo, 2020; Torelli,

2021).

First, by integrating the Sociology of Worth (Boltanski et al., 2006; Boltanski and Thévenot, 1991) with ethical-religious theory, the study advances a multi-dimensional account of organisational justification. The identification of distinct “worlds of value” as discursive repertoires through which firms construct legitimacy demonstrates that sustainability reporting is not merely a site of information transmission, but a space of moral evaluation and normative narration. Reports function as justificatory devices, aligning corporate actions with culturally resonant value logics. This challenges technocratic and instrumentalist views of disclosure and positions reporting as an ethical practice.

Second, the comparative analysis between Italy and the UK shows how Catholic and Protestant ethics may shape the selection, prioritisation, and articulation of value worlds. Catholic-inspired contexts tend to favour narratives grounded in solidarity, institutional trust, and social equity, while Protestant-influenced companies tend to emphasise efficiency, individual accountability, and performance. This finding adds to the literature on religion and corporate behaviour (Liao et al., 2019; Minton et al., 2019) by demonstrating that religiosity operates not only at the level of individual moral orientation or governance structures, but also through the discursive construction of organisational legitimacy within corporate reporting practices.

Third, the study refines the application of the SOW framework by contextualising it within sector-specific and culturally embedded discourses. By focusing on the water utility sector—a domain inherently linked to public value, environmental sustainability, and social rights—the research demonstrates that not all worlds of worth are equally available or institutionally viable across settings. The varying salience of the Civic and Domestic worlds, for example, underscores the need to consider how sectoral logics and governance arrangements mediate the ethical repertoire available to organisations.

Finally, the findings invite a reconsideration of how materiality is conceptualised in sustainability reporting. Although this study does not examine formal materiality assessment processes in corporate reports, the analysis shows that sustainability issues are framed and justified through ethical repertoires that vary across contexts. This suggests that what comes to be regarded as “material” may also be shaped by the normative frameworks through which organisations interpret their responsibilities and societal roles. In this sense, the study opens new avenues for theorising materiality as a socially constructed and culturally contingent category, challenging the assumption that standardisation alone can ensure relevance and comparability across diverse institutional contexts.

6.2. Practical and policy implications

From a practical perspective, the study offers several implications for regulators, companies, and standard-setters concerned with enhancing the quality, credibility, and cultural responsiveness of sustainability reporting.

First, for report preparers and managers, the research highlights the importance of engaging more explicitly with the ethical foundations of their disclosures. Rather than treating sustainability as a matter of compliance or reputation management, organisations - particularly in sectors providing essential public services - may enhance the legitimacy of their communication by rooting it in locally resonant ethical narratives. For Italian utilities, this may mean emphasising the social embeddedness and redistributive function of their mission; for UK firms, it may involve a more reflexive articulation of the values underpinning performance metrics and environmental targets.

Second, for regulators and public authorities, the findings underline the need to recognise the role of cultural and institutional diversity in shaping the interpretation and reception of sustainability reports. Regulatory frameworks that impose highly standardised disclosure requirements may overlook the ethical and symbolic work that reporting performs in different contexts. Allowing some flexibility in narrative

structure - particularly in the articulation of purpose, value, and legitimacy - could allow companies to reflect more authentically the moral commitments they seek to express. This is especially pertinent in the context of emerging EU regulations (e.g., the European Sustainability Reporting Standards) and global initiatives (e.g., the IFRS Sustainability Disclosure Standards), which must navigate tensions between comparability and contextual relevance. These considerations are particularly relevant in regulated utility sectors such as water services, where sustainability reporting increasingly interacts with public oversight, regulatory accountability, and debates on the governance of essential services.

Third, for investors and stakeholders, the study invites a more nuanced reading of sustainability reports. Beyond performance indicators and ESG scores, stakeholders should attend to the justificatory logics that underpin disclosures, asking questions such as: what kinds of value are being promoted? Whose interests are being prioritised? What ethical commitments are implicit in the structure of the report? Understanding these dimensions can help stakeholders distinguish between performative compliance and substantive accountability.

Finally, the findings carry implications for international standard-setters. As the demand for global reporting frameworks grows, this study highlights the limits of one-size-fits-all approaches. The emergence of distinct ethical-narrative regimes across countries suggests that standards may need to remain dialogical and open to pluralism, incorporating mechanisms that allow for ethical localisation. This may involve providing guidance on cultural adaptation, encouraging the disclosure of value assumptions, or supporting context-specific indicators that reflect diverse visions of sustainability.

6.3. Limitations

While this study offers original insights into the ethical and cultural foundations of sustainability reporting, several limitations must be acknowledged, which also open avenues for future research.

First, the analysis is limited to a single sector, namely water utilities. Although this sector is particularly relevant for the study of public value and environmental stewardship, its specific institutional and regulatory characteristics may not be representative of other industries. The public service mission, infrastructural nature, and regulatory oversight typical of the water sector may amplify certain ethical registers, particularly the Civic and Green Worlds, which may be less prominent in more commercially oriented sectors. Future research could therefore examine whether similar ethical patterns emerge in industries with different governance structures or stakeholder configurations, such as energy, finance, or manufacturing.

Second, while the comparative design between the UK and Italy enables a culturally grounded analysis, the focus on these two national contexts, broadly associated with Protestant and Catholic traditions, does not capture the full diversity of ethical and religious influences that may shape corporate reporting. Other institutional and cultural settings, including Orthodox, Islamic, or more secular contexts, may produce alternative configurations of value, legitimacy, and moral justification. Extending the analysis to a broader range of countries would therefore strengthen the theoretical robustness and external validity of the findings.

Third, the study focuses on sustainability reports as discursive artefacts, privileging narrative content and justificatory structures. While this approach is theoretically fruitful, it does not capture how these narratives are received, interpreted, or contested by stakeholders. Future studies could therefore complement textual analysis with stakeholder interviews or perception-based methods to explore the resonance, credibility, or contestation of the values articulated in corporate reports.

Fourth, the methodological reliance on qualitative content analysis, although systematically structured, inevitably involves a degree of interpretive judgement. Despite efforts to enhance reliability through

intercoder agreement, the categorisation of textual elements into worlds of value may remain sensitive to contextual nuances and interpretive framing. In addition, the analysis relies on a single reporting year for each country, which limits the possibility of examining how sustainability narratives evolve. Future research could extend the temporal scope by incorporating multiple reporting cycles and exploring how value-world narratives respond to regulatory changes and shifting societal expectations. Another promising direction would be to investigate the role of ownership structures, particularly differences between publicly listed, privately owned, and publicly controlled utilities, which may introduce additional variation in reporting practices and justificatory logics.

6.4. Concluding remarks

Despite these limitations, the study makes a series of substantive contributions to the literature on sustainability reporting, business ethics, and critical accounting. First, it offers a novel integration of the Sociology of Worth with religious ethics, advancing a theoretical framework that captures the normative diversity underpinning sustainability discourses. By demonstrating how ethical-religious traditions may shape the mobilisation of different worlds of value, the study moves beyond functionalist or instrumentalist interpretations of reporting and positions disclosure as a moral and culturally embedded practice.

Second, the study contributes to comparative sustainability scholarship by highlighting how context-specific ethical infrastructures shape the articulation of corporate legitimacy. The contrasting prominence of the Market, Civic, Domestic, and Green Worlds across UK and Italian reports illustrates that sustainability is not a universally shared concept but a culturally mediated construct enacted through distinct justificatory grammars. This insight has important implications for the design of global reporting standards and their sensitivity to normative pluralism.

Third, by focusing on value narratives rather than performance metrics, the study enriches the understanding of how organisations construct meaning around sustainability. It suggests that legitimacy in non-financial disclosure is not produced solely through alignment with standards or data completeness, but also through the ethical coherence and cultural intelligibility of the narrative itself. This finding invites scholars and practitioners to pay closer attention to the rhetorical, ethical, and symbolic dimensions of corporate reporting, addressing an important gap in existing research that has predominantly focused on the extent or credibility of disclosure rather than on its underlying justificatory logics.

In conclusion, this study argues that sustainability reporting should be understood not merely as a technical or regulatory function, but as an ethically situated and culturally contingent form of communication. By tracing how Catholic and Protestant traditions interact with institutional contexts to shape the value worlds mobilised by water utilities in Italy and the UK, the study provides a theoretically grounded account of the moral architectures underpinning sustainability disclosure. In doing so, it invites a broader rethinking of what it means for organisations to be accountable, not only in terms of what they report, but also in terms of how and why they construct legitimacy through ethically resonant narratives. Such a perspective is essential for developing more pluralistic and meaningful approaches to corporate sustainability in an increasingly complex and normatively fragmented global context.

CRedit authorship contribution statement

Gabriella D'Amore: Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Formal analysis, Conceptualization. **Antonio D'Alessio:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alessandra Di Fraia:** Data curation. **Fabio Serini:** Writing – original draft. **Loris Landriani:** Writing – review & editing, Writing – original draft, Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Agovino, M., Cerciello, M., Garofalo, A., Landriani, L., Lepore, L., 2021. Corporate governance and sustainability in water utilities. The effects of decorporatisation in the city of Naples, Italy. *Bus. Strat. Environ.* 30 (2), 874–890. <https://doi.org/10.1002/bse.2659>.
- Aji, H.M., Sutikno, B., 2015. The extended consequence of greenwashing: perceived consumer skepticism. *International Journal of Business and Information, Academy of Taiwan Information Systems Research* 10 (4), 433.
- Akhmouch, A., Correia, F.N., 2016. The 12 OECD principles on water governance—when science meets policy. *Util. Policy* 43, 14–20. Elsevier.
- Alshehhi, A., Nobanee, H., Khare, N., 2018. The impact of sustainability practices on corporate financial performance: literature trends and future research potential. *Sustainability, Multidisciplinary Digital Publishing Institute* 10 (2), 494.
- Arnone, M., Leogrande, A., Drago, C., Costantiello, A., 2024. Social trust and support networks: a regional analysis of Italy. Preprints. doi: 10.20944/preprints202409.1203.v1.
- Arrunada, B., 2010. Protestants and Catholics: similar work ethic, different social ethic. *Econ. J.* 120 (547), 890–918. Oxford University Press Oxford, UK.
- Baker, M., Gray, R., Schaltegger, S., 2023. Debating accounting and sustainability: from incompatibility to rapprochement in the pursuit of corporate sustainability. *Account Audit. Account. J.* 36 (2), 591–619. Emerald Publishing Limited.
- Bakker, S., 2014. "Actor rationales in sustainability transitions—Interests and expectations regarding electric vehicle recharging". *Environ. Innov. Soc. Transit.* 13, 60–74. Elsevier.
- Baldini, M., Maso, L.D., Liberatore, G., Mazzi, F., Terzani, S., 2018. Role of country-and firm-level determinants in environmental, social, and governance disclosure. *J. Bus. Ethics* 150 (1), 79–98. Springer.
- Barbosa, A., Brusca, I., 2015. "Governance structures and their impact on tariff levels of Brazilian water and sanitation corporations". *Util. Policy* 34, 94–105. Elsevier.
- Bartoletti, R., Faccioli, F., 2016. Public engagement, local policies, and citizens' participation: an Italian case study of civic collaboration. *Soc. Media Soc.* 2 (3), 2056305116662187. <https://doi.org/10.1177/2056305116662187>.
- Battisti, G., Stoneman, P., 2010. How innovative are UK firms? Evidence from the fourth UK community innovation survey on synergies between technological and organizational innovations. *Br. J. Manag.* 21 (1), 187–206. <https://doi.org/10.1111/j.1467-8551.2009.00629.x>.
- Beal, C.D., Flynn, J., 2015. "Toward the digital water age: survey and case studies of Australian water utility smart-metering programs". *Util. Policy* 32, 29–37. Elsevier.
- Bebbington, J., Larrinaga-González, C., 2008. Carbon trading: accounting and reporting issues. *Eur. Account. Rev.* 17 (4), 697–717. <https://doi.org/10.1080/09638180802489162>.
- Bel, G., Warner, M., 2008. Does privatization of solid waste and water services reduce costs? A review of empirical studies. *Resour. Conserv. Recycl.* 52 (12), 1337–1348. Elsevier.
- Bhattacharya, A., Agbola, F.W., 2018. Social and environmental reporting and the co-creation of corporate legitimacy. *Contemp. Manag. Res.* 14 (3), 191–223.
- Bijani, M., Hayati, D., Azadi, H., Tanaskovic, V., Witlox, F., 2020. Causes and consequences of the conflict among agricultural water beneficiaries in Iran. *Sustainability* 12 (16), 6630. MDPI.
- Boeri, A., Longo, D., Gianfrate, V., Lorenzo, V., 2017. Resilient communities. Social infrastructures for sustainable growth of urban areas. A case study. *Int. J. Sustain. Dev. Plann.* 12 (2), 227–237. WIT Press.
- Boiral, O., 2013. Sustainability reports as simulacra? A counter-account of A and A+ GRI reports. *Account Audit. Account. J.* 26 (7), 1036–1071. Emerald Group Publishing Limited.
- Boltanski, L., Thévenot, L., 1991. *De La Justification: Les Économies De La Grandeur*. Boltanski, L., Thvenot, L., Porter, C., 2006. *On Justification: Economies of Worth*. Princeton University Press. <https://doi.org/10.2307/j.ctv1zm2tzm>.
- Brammer, S., Williams, G., Zinkin, J., 2007. Religion and attitudes to corporate social responsibility in a large cross-country sample. *J. Bus. Ethics* 71 (3), 229–243. Springer.
- Brunelli, S., Di Carlo, E. (Eds.), 2020. *Accountability, Ethics and Sustainability of Organizations: New Theories, Strategies and Tools for Survival and Growth*. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-030-31193-3>.
- Bruni, L., 2018. Capitalism and its new-old religion: a civil economy perspective. *Journal for Markets and Ethics* 6 (1), 121–131. Sciendo.
- Caputo, F., Ligorio, L., Pizzi, S., 2021. The contribution of higher education institutions to the SDGs—An evaluation of sustainability reporting practices. *Adm. Sci.* 11 (3), 97. MDPI.

- Casadevall, S.R., 2016. "Improving the management of water multi-functionality through stakeholder involvement in decision-making processes". *Util. Policy* 43, 71–81. Elsevier.
- Chantziaras, A., Dedoulis, E., Grougiou, V., Leventis, S., 2020. "The impact of religiosity and corruption on CSR reporting: the case of US banks". *J. Bus. Res.* 109, 362–374. Elsevier.
- Cherchi, C., Braduzzaman, M., Oppenheimer, J., Bros, C.M., Jacangelo, J.G., 2015. "Energy and water quality management systems for water utility's operations: a review". *J. Environ. Manag.* 153, 108–120. Elsevier.
- Christie, I., Gunton, R.M., Hejnowicz, A.P., 2019. Sustainability and the common good: Catholic Social Teaching and 'Integral Ecology' as contributions to a framework of social values for sustainability transitions. *Sustain. Sci.* 14 (5), 1343–1354.
- Cifolelli, S., Ziruolo, A., Berardi, M., 2024. Creating public value within the smart energy communities (SECs): a theoretical framework for hybrid organizations. *International Public Management Review* 24 (1), 46–70.
- Cloutier, C., Langley, A., 2013. The logic of institutional logics: Insights from French pragmatist sociology. *J. Manag. Inq.* 22 (4), 360–380.
- Contrafatto, M., 2014. Stewardship theory: approaches and perspectives. In: Costa, E., Parker, L.D., Andreas, M. (Eds.), *Advances in Public Interest Accounting*, 17. Emerald Group Publishing Limited, pp. 177–196. <https://doi.org/10.1108/S1041-70602014000017007>.
- Correia, T., Marques, R.C., 2011. Performance of Portuguese water utilities: how do ownership, size, diversification and vertical integration relate to efficiency? *Water Policy* 13 (3), 343–361. IWA Publishing.
- Cunha Marques, R., 2008. Comparing private and public performance of Portuguese water services. *Water Policy* 10 (1), 25–42. IWA Publishing.
- D'Amore, G., Landriani, L., Lepore, L., 2021. Ownership and sustainability of Italian water utilities: the stakeholder role. *Util. Policy* 71, 101228.
- D'Amore, G., Di Vaio, A., Balsalobre-Lorente, D., Boccia, F., 2022. "Artificial intelligence in the water-energy-food model: a holistic approach towards sustainable development goals". *Sustainability* 14 (2), 867. MDPI.
- D'Amore, G., Testa, M., Lepore, L., 2024. "How Is the Utilities Sector Contributing to Building a Sustainable Future? A Systematic Literature Review of Sustainability Practices". *Sustainability* 16 (1), 374. <https://doi.org/10.3390/su16010374>. Multidisciplinary Digital Publishing Institute.
- De Villiers, C., Marques, A., 2016. Corporate social responsibility, country-level predispositions, and the consequences of choosing a level of disclosure. *Account. Bus. Res.* 46 (2), 167–195. <https://doi.org/10.1080/00014788.2015.1039476>.
- De Villiers, C., Van Staden, C.J., 2011. "Where firms choose to disclose voluntary environmental information". *J. Account. Publ. Pol.* 30 (6), 504–525. Elsevier.
- Delmas, M.A., Burbano, V.C., 2011. The drivers of greenwashing. *Calif. Manag. Rev.* 54 (1), 64–87. <https://doi.org/10.1525/cmr.2011.54.1.64>.
- Di Vaio, A., Trujillo, L., D'Amore, G., Palladino, R., 2021. Water governance models for meeting sustainable development goals: a structured literature review. *Util. Policy* 72, 101255. Elsevier.
- Earle, T.C., Cvetkovich, G., 2013. Social trust and culture in risk management. In: *Social Trust and the Management of Risk*. Routledge, pp. 9–21.
- Fabbri, P., Fraquelli, G., 2000. Costs and structure of technology in the Italian water industry. *Empirica* 27 (1), 65–82. Springer.
- Farooq, Q., Hao, Y., Liu, X., 2019. Understanding corporate social responsibility with cross-cultural differences: a deeper look at religiosity. *Corp. Soc. Responsib. Environ. Manag.* 26 (4), 965–971. <https://doi.org/10.1002/csr.1736>.
- Fassin, Y., Van Rossem, A., Buelens, M., 2011. Small-business owner-managers' perceptions of business ethics and CSR-related concepts. *J. Bus. Ethics* 98 (3), 425–453. Springer.
- Forman, G., 2003. An extensive empirical study of feature selection metrics for text classification. *J. Mach. Learn. Res.* 3 (Mar), 1289–1305.
- Francis, P., 2015. *Laudato Si: On Care for Our Common Home*. Our Sunday Visitor. Huntington, IN.
- Freeman, R.E., 2023. The politics of stakeholder theory: some future directions. In: Dmytriiev, S.D., Freeman, R.E., Edward, R. (Eds.), *Freeman's Selected Works on Stakeholder Theory and Business Ethics*, 53. Springer International Publishing, Cham, pp. 119–132. https://doi.org/10.1007/978-3-031-04564-6_5.
- Freudenburg, W.R., 1993. Risk and recreancy: weber, the division of labor, and the rationality of risk perceptions. *Soc. Forces* 71 (4), 909–932. The University of North Carolina Press.
- Friedman, A.L., Miles, S., 2001. "Socially responsible investment and corporate social and environmental reporting in the UK: an exploratory study". *Br. Account. Rev.* 33 (4), 523–548. Elsevier.
- Furlong, C., Jegatheesan, J., Currell, M., Iyer-Raniga, U., Khan, T., Ball, A.S., 2019. "Is the global public willing to drink recycled water? A review for researchers and practitioners". *Util. Policy* 56, 53–61. Elsevier.
- Garde Sánchez, R., Rodríguez Bolívar, M.P., López Hernández, A.M., 2017. Corporate and managerial characteristics as drivers of social responsibility disclosure by state-owned enterprises. *Rev. Manag. Sci.* 11 (3), 633–659. Springer.
- Gray, R., 2010. Is accounting for sustainability actually accounting for sustainability... and how would we know? An exploration of narratives of organisations and the planet. *Account. Org. Soc.* 35 (1), 47–62. <https://doi.org/10.1016/j.aos.2009.04.006>.
- Guerrini, A., Romano, G., Leardini, C., 2018. "Economies of scale and density in the Italian water industry: a stochastic frontier approach". *Util. Policy* 52, 103–111. Elsevier.
- Guiso, L., Sapienza, P., Zingales, L., 2006. Does culture affect economic outcomes? *J. Econ. Perspect.* 20 (2), 23–48. American Economic Association.
- Gunnarsdottir, M.J., Gardarsson, S.M., Elliott, M., Sigmundsdottir, G., Bartram, J., 2012. Benefits of water safety plans: microbiology, compliance, and public health. *Environmental Science & Technology* 46 (14), 7782–7789. <https://doi.org/10.1021/es300372h>.
- Hahn, T., Figge, F., Pinkse, J., Preuss, L., 2010. Trade-offs in corporate sustainability: you can't have your cake and eat it. *Bus. Strat. Environ.* 19 (4), 217–229. <https://doi.org/10.1002/bse.674>.
- Harper, D., Thompson, A.R. (Eds.), 2011. *Qualitative Research Methods in Mental Health and Psychotherapy: a Guide for Students and Practitioners*, first ed. Wiley. <https://doi.org/10.1002/9781119973249>.
- Hollenbach, D., 2002. *The Common Good and Christian Ethics*, 22. Cambridge university press.
- Hrudey, S.E., Hrudey, E.J., Pollard, S.J., 2006. "Risk management for assuring safe drinking water". *Environ. Int.* 32 (8), 948–957. Elsevier.
- Hu, Z., Morton, L.W., Mahler, R.L., 2011. "Bottled water: united States consumers and their perceptions of water quality". *Int. J. Environ. Res. Publ. Health* 8 (2), 565–578. Molecular Diversity Preservation International (MDPI).
- Hunter, J.D., 1987. *Evangelicalism: The Coming Generation*. University of Chicago Press, Chicago.
- Imperiale, F., Pizzi, S., Lippolis, S., 2023. "Sustainability reporting and ESG performance in the utilities sector". *Util. Policy* 80, 101468. Elsevier.
- Jagd, S., 2011. Pragmatic sociology and competing orders of worth in organizations. *Eur. J. Soc. Theory* 14 (3), 343–359.
- Jiménez, A., Saikia, P., Giné, R., Avello, P., Leten, J., Liss Lymer, B., Schneider, K., et al., 2020. Unpacking water governance: a framework for practitioners. *Water* 12 (3), 827. MDPI.
- Kim, W., Kim, E., Song, W., 2025. Evaluating the cooling benefits of rainwater spraying in urban environments using machine learning and UAV thermal imaging. *Landsc. Ecol. Eng.* <https://doi.org/10.1007/s11355-025-00662-9>.
- Koehrsen, J., 2015. Does religion promote environmental sustainability? – exploring the role of religion in local energy transitions. *Soc. Compass* 62 (3), 296–310. <https://doi.org/10.1177/0037768615587808>.
- Krippendorff, K., 1980. Validity in content analysis. *Computerstrategien Für Die Kommunikationsanalyse* 291, 69–112.
- Krishnan, S.R., Nallakuruppan, M.K., Chengoden, R., Koppu, S., Iyapparaja, M., Sadhasivam, J., Sethuraman, S., 2022. Smart water resource management using artificial intelligence—A review. *Sustainability* 14 (20), 13384. MDPI.
- Kuckartz, U., Rädiker, S., 2019. Introduction: analyzing qualitative data with software. In: *Analyzing Qualitative Data with MAXQDA*. Springer International Publishing, Cham, pp. 1–11. https://doi.org/10.1007/978-3-030-15671-8_1.
- Levallois, P., Grondin, J., Gingras, S., 1999. Evaluation of consumer attitudes on taste and tap water alternatives in Quebec. *Water Sci. Technol.* 40 (6), 135–139. Elsevier.
- Liao, Z., Dong, J., Weng, C., Shen, C., 2019. CEOs' religious beliefs and the environmental innovation of private enterprises: the moderating role of political ties. *Corp. Soc. Responsib. Environ. Manag.* 26 (4), 972–980. <https://doi.org/10.1002/csr.1737>.
- Ligorio, L., Venturelli, A., Caputo, F., 2025. Driving SDGs contribution: governance impact in state-owned vs. private companies' CSR reporting. *Soc. Responsib. J.* 21 (3), 451–472. Emerald Publishing Limited.
- Lima, S., Brochado, A., Marques, R.C., 2021. Public-private partnerships in the water sector: a review. *Util. Policy* 69, 101182. Elsevier.
- Liu, L., Jensen, M.B., 2018. "Green infrastructure for sustainable urban water management: practices of five forerunner cities". *Cities* 74, 126–133. Elsevier.
- Lombardi, G.V., Stefani, G., Paci, A., Becagli, C., Miliacca, M., Gastaldi, M., Giannetti, B. F., et al., 2019. "The sustainability of the Italian water sector: an empirical analysis by DEA". *J. Clean. Prod.* 227, 1035–1043. Elsevier.
- Mahmood, Z., Uddin, S., 2021. Institutional logics and practice variations in sustainability reporting: evidence from an emerging field. *Account. Audit. Account. J.* 34 (5), 1163–1189. Emerald Publishing Limited.
- Manes-Rossi, F., Nicolo', G., 2022. Exploring sustainable development goals reporting practices: from symbolic to substantive approaches—Evidence from the energy sector. *Corp. Soc. Responsib. Environ. Manag.* 29 (5), 1799–1815. <https://doi.org/10.1002/csr.2328>.
- Marin, P., 2009. *Public Private Partnerships for Urban Water Utilities: a Review of Experiences in Developing Countries*. World Bank Publications.
- Maroun, W., Ecim, D., 2024. Biodiversity reporting by United Kingdom (UK)-listed companies: a review of extent, content and readability of disclosures. *Bus. Strat. Environ.* 33 (8), 7800–7824. <https://doi.org/10.1002/bse.3866>.
- Massaro, T., 2023. *Living Justice: Catholic Social Teaching in Action*. Bloomsbury Publishing, New York.
- McCleary, R.M., 2007. Salvation, damnation, and economic incentives. *J. Contemp. Relig.* 22 (1), 49–74. <https://doi.org/10.1080/13537900601114503>.
- Michelon, G., Parbonetti, A., 2012. The effect of corporate governance on sustainability disclosure. *J. Manag. Govern.* 16 (3), 477–509. <https://doi.org/10.1007/s10997-010-9160-3>.
- Minton, E.A., Johnson, K.A., Liu, R.L., 2019. "Religiosity and special food consumption: the explanatory effects of moral priorities". *J. Bus. Res.* 95, 442–454. Elsevier.
- Mio, C., Fasan, M., Scarpa, F., 2023. "Materiality investor perspectives on utilities' ESG performance. An empirical analysis of ESG factors and cost of equity". *Util. Policy* 82, 101555. Elsevier.
- Moore, M.H., 1997. *Creating Public Value: Strategic Management in Government*. Harvard university press.
- Morris, J., Sassen, R., McGuinness, M., 2023. Beyond water scarcity and efficiency? Water sustainability disclosures in corporate reporting. *Sustain. Account. Manag. Policy J.* 14 (3), 490–514. Emerald Publishing Limited.
- Moussa, A.S., Elmarzouky, M., 2024. Beyond compliance: how ESG reporting influences the cost of capital in UK firms. *J. Risk Financ. Manag.* 17 (8), 326.

- Nandi, S., Gonela, V., 2022. Rainwater harvesting for domestic use: a systematic review and outlook from the utility policy and management perspectives. *Util. Policy* 77, 101383. Elsevier.
- Neuendorf, K.A., 2017. *The Content Analysis Guidebook*. sage.
- Nti, I.K., Adekoya, A.F., Weyori, B.A., Nyarko-Boateng, O., 2022. Applications of artificial intelligence in engineering and manufacturing: a systematic review. *J. Intell. Manuf.* 33 (6), 1581–1601. <https://doi.org/10.1007/s10845-021-01771-6>.
- O'Callaghan, P., Adapa, L.M., Buisman, C., 2020. How can innovation theories be applied to water technology innovation? *J. Clean. Prod.* 276, 122910. Elsevier.
- O'Dochartaigh, A., 2019. No more fairytales: a quest for alternative narratives of sustainable business. *Account. Audit. Account. J.* 32 (5), 1384–1413. Emerald Publishing Limited.
- O'Dwyer, B., Unerman, J., 2007. From functional to social accountability: transforming the accountability relationship between funders and non-governmental development organisations. *Account. Audit. Account. J.* 20 (3), 446–471. Emerald Group Publishing Limited.
- Patriotta, G., Gond, J., Schultz, F., 2011. Maintaining legitimacy: controversies, orders of worth, and public justifications. *J. Manag. Stud.* 48 (8), 1804–1836. <https://doi.org/10.1111/j.1467-6486.2010.00990.x>.
- Pérez-Urdiales, M., Libra, J.M., Machado, K.B., Serebrisky, T., Sosa, B.S., 2024. Household water bill perception in Brazil. *Util. Policy* 87, 101704. Elsevier.
- Pinto, F.S., Marques, R.C., 2015. Tariff structures for water and sanitation urban households: a primer. *Water Policy* 17 (6), 1108–1126. IWA Publishing.
- Pizzi, S., Caputo, F., Venturelli, A., 2020. Accounting to ensure healthy lives: critical perspective from the Italian national healthcare system. *Corporate Governance: The International Journal of Business in Society* 20 (3), 445–460. Emerald Publishing Limited.
- Pontifical Council for Justice and Peace, 2004. *Compendium of the Social Doctrine of the Church*. Vatican City: Libreria Editrice Vaticana.
- Putnam, R.D., 2000. *Bowling Alone: The Collapse and Revival of American Community*. Simon and Schuster, New York.
- Ramírez-Agudelo, N.A., de Pablo, J., Roca, E., 2021. "Exploring alternative practices in urban water management through the lens of circular economy—A case study in the Barcelona metropolitan area". *J. Clean. Prod.* 329, 129565. Elsevier.
- Remo-Diez, N., Mendana-Cuervo, C., Arenas-Parra, M., 2023. Exploring the asymmetric impact of sustainability reporting on financial performance in the utilities sector: a longitudinal comparative analysis. *Util. Policy* 84, 101650. Elsevier.
- Roeger, A., Tavares, A.F., 2018. Water safety plans by utilities: a review of research on implementation. *Util. Policy* 53, 15–24. Elsevier.
- Rogers, P., Hall, A.W., 2003. *Effective Water Governance*, 7. Global water partnership, Stockholm.
- Romano, G., Guerrini, A., 2011. "Measuring and comparing the efficiency of water utility companies: a data envelopment analysis approach". *Util. Policy* 19 (3), 202–209. Elsevier.
- Romano, O., Akhmouch, A., 2019. Water governance in cities: current trends and future challenges. *Water* 11 (3), 500. MDPI.
- Rousseau, H.E., 2017. Corporate sustainability: toward a theoretical integration of catholic social teaching and the natural-resource-based view of the firm. *J. Bus. Ethics* 145 (4), 725–737. Springer.
- Rowe, W.L., 2007. *William L. Rowe on Philosophy of Religion: Selected Writings*. Ashgate Publishing, Ltd.
- Rowe, T.S., 2017. *Toward a Better Worldliness: Ecology, Economy, and the Protestant Tradition*. Fortress Press, Minneapolis.
- Sherkat, D.E., Ellison, C.G., 2007. Structuring the religion-environment connection: identifying religious influences on environmental concern and activism. *J. Sci. Stud. Relig.* 46 (1), 71–85. <https://doi.org/10.1111/j.1468-5906.2007.00341.x>.
- Siano, A., Voller, A., Conte, F., Amabile, S., 2017. 'More than words': expanding the taxonomy of greenwashing after the Volkswagen scandal". *J. Bus. Res.* 71, 27–37. Elsevier.
- Silva, S., 2021. Corporate contributions to the sustainable development goals: an empirical analysis informed by legitimacy theory. *J. Clean. Prod.* 292, 125962. Elsevier.
- Silvestre, H.C., Marques, R.C., Gomes, R.C., 2018. Joined-up government of utilities: a meta-review on a public-public partnership and inter-municipal cooperation in the water and wastewater industries. *Public Manag. Rev.* 20 (4), 607–631. <https://doi.org/10.1080/14719037.2017.1363906>.
- Taiwo, R., Ben Seghier, M.E.A., Zayed, T., 2023. Toward sustainable water infrastructure: the state-of-the-art for modeling the failure probability of water pipes. *Water Resour. Res.* 59 (4), 33256. <https://doi.org/10.1029/2022WR033256>.
- Talbot, D., Boiral, O., 2018. GHG reporting and impression management: an assessment of sustainability reports from the energy sector. *J. Bus. Ethics* 147 (2), 367–383. <https://doi.org/10.1007/s10551-015-2979-4>.
- Tawney, R.H., 1926. *Religion and the Rise of Capitalism: A Historical Study*. Harcourt Brace, New York.
- Terzani, S., Turzo, T., 2021. Religious social norms and corporate sustainability: the effect of religiosity on environmental, social, and governance disclosure. *Corp. Soc. Responsib. Environ. Manag.* 28 (1), 485–496. <https://doi.org/10.1002/csr.2063>.
- Terzani, S., Turzo, T., Moretti, G., 2025. Environmental reporting and religiosity: an empirical analysis within the European financial sector. *Financ. Report. FrancoAngeli Editore*, No. 2025/1, 117–147. doi: 10.3280/fr202516716.
- Tommasetti, A., Mussari, R., Maione, G., Sorrentino, D., 2020. Sustainability accounting and reporting in the public sector: towards public value co-creation? *Sustainability* 12 (5), 1909. MDPI.
- Tonnarelli, F., Vermiglio, C., Migliardo, C., Naciti, V., 2025. The impact of EU taxonomy for sustainable activities on European utilities' performance. *Bus. Strat. Environ.* 34 (3), 2848–2862. <https://doi.org/10.1002/bse.4128>.
- Torelli, R., 2021. Sustainability, responsibility and ethics: different concepts for a single path. *Soc. Responsib. J.* 17 (5), 719–739. Emerald Publishing Limited.
- Torelli, R., Balluchi, F., Lazzini, A., 2020. Greenwashing and environmental communication: effects on stakeholders' perceptions. *Bus. Strat. Environ.* 29 (2), 407–421. <https://doi.org/10.1002/bse.2373>.
- Traxler, A.A., Greiling, D., 2019. Sustainable public value reporting of electric utilities. *Baltic J. Manag.* 14 (1), 103–121. Emerald Publishing Limited.
- Tregidga, H., Milne, M.J., 2006. From sustainable management to sustainable development: a longitudinal analysis of a leading New Zealand environmental reporter. *Bus. Strat. Environ.* 15 (4), 219–241. <https://doi.org/10.1002/bse.534>.
- Tsatsou, A., Pergar, P., Frantzeskaki, N., Malamis, S., Atanasova, N., 2023. Planning nature-based solutions for water management and circularity in Ljubljana, Slovenia: examining how urban practitioners navigate barriers and perceive institutional readiness. *Urban For. Urban Green.* 89, 128090. Elsevier.
- Veltri, S., Bruni, M.E., Iazzolino, G., Morea, D., Baldissarro, G., 2023. "Do ESG factors improve utilities corporate efficiency and reduce the risk perceived by credit lending institutions? An empirical analysis". *Util. Policy* 81, 101520. Elsevier.
- Venturelli, A., Ligorio, L., de Nuccio, E., 2023. Biodiversity accountability in water utilities: a case study. *Util. Policy* 81, 101495. Elsevier.
- Walter, M., Cullmann, A., von Hirschhausen, C., Wand, R., Zschille, M., 2009. Quo vadis efficiency analysis of water distribution? A comparative literature review. *Util. Policy* 17 (3–4), 225–232. Elsevier.
- Weber, M., 2003. *The Protestant Ethic and the Spirit of Capitalism*, Translated by Parsons. Dover Pubns, Mineola, N.Y.
- Weisner, M.L., Root, T.L., Harris, M.S., Mitsova, D., Liu, W., 2020. The complexities of trust between urban water utilities and the public. *Sustain. Water Resour. Manag.* 6 (3), 50. <https://doi.org/10.1007/s40899-020-00407-6>.
- Westling, E.L., Sharp, L., Scott, D., Tait, S., Rychlewski, M., Ashley, R.M., 2019. Reflexive adaptation for resilient water services: lessons for theory and practice. *Glob. Environ. Change* 57, 101937. Elsevier.
- Yongvanich, K., Guthrie, J., 2007. Legitimation strategies in Australian mining extended performance reporting. *Journal of Human Resource Costing & Accounting* 11 (3), 156–177. Emerald Group Publishing Limited.
- Zheng, Z., Wu, X., Srihari, R., 2004. Feature selection for text categorization on imbalanced data. *ACM SIGKDD Explorations Newsletter* 6 (1), 80–89. <https://doi.org/10.1145/1007730.1007741>.